Analytical Characterization of Multi-State Effective Discharge Rates for Bus-Only Lane Conversion Scheduling Problem

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Abstract

An accurate quantification of traffic flow characteristics with and without bus-only lane is important for determining transit priority strategies. Examinations of real-world vehicle trajectory data, like those from the NGSIM dataset, found that an effective discharge rate may be discounted when merging or lane-changing behaviors are observed. To deal with this, this manuscript analytically derives the effective discharge rate of a roadway segment by considering multi-stage merging behavior and vehicular traffic kinematics near a side entrance, for optimal bus-only lane conversion. The classic fluid-based approximation model by Newell that characterizes the queuing and dissipation process is extended to a multi-state queuing analysis framework. Three effective discharge rate discount factors are explicitly derived, for bus-only lane, general-purpose lane, and mixed-traffic lane, respectively. The derived effective discharge rates are represented by mathematically-simple expressions in a closed-form, and correctly account for the effects of demand inputs, such as arrival rate and the ratio of vehicles of different types, vehicular performance (such as acceleration rate), and traffic flow characteristics (such as backward wave speed and jam density). The bus-only lane conversion scheduling problem is then formulated as an optimization model to minimize total traffic delay. Validation, using NGSIM data, showed that our model reduced the effective discharge rate estimation error significantly on both arterials and freeways. Sensitivity analysis revealed that the optimal bus-only lane scheduling time varied in response to traffic demand and vehicle ratios, and the developed optimization model was always beneficial when compared with the demand-oriented strategy.

Keyword: Effective discharge rate; bus-only lane; optimal scheduling; queuing profile; shockwave propagation;

1. Introduction

A bus-only lane comes at the cost of reduced car-carrying capacity, which may lead to a long queue and excessive delay of general vehicles. An accurate quantification of traffic flow characteristics, with and without a bus-only lane, is important for determining transit priority strategies. Following classic traffic flow fundamentals, traffic dynamics have long been modeled as the outcomes of demand-supply interactions, with demand being time varying, whereas supply takes a fixed capacity value (i.e., effective discharge rate). The vehicle queueing process is usually formulated as the integral of the difference between the arrival flow rate and the fixed capacity value. For example, in the morning commute problem (Vickrey 1969), the effective discharge rate at a bottleneck was assumed to be constant, so that the queue discharge time could be derived with a simple form. Almost all subsequent research follows such an assumption (Newell 1987, Arnott et al. 1990, Daganzo and Garcia 2000). In another example, based on a simplified car-following model (Newell 2002), a triangular fundamental diagram was derived, in which the maximum discharge rate and speed of the backward wave could be uniquely identified, once the microscopic car-following behavior was calibrated. Such an approach and assumption were adopted in numerous later research studies such as Ahn et al. (2004), Chen et al. (2012), Wei et al. (2017), and many others. A similar assumption is also applied to the bus-only lane conversion scheduling problem. For example, Saade et al. (2018) used a simulation-based approach to study what could occur when select street lanes are periodically reserved for buses throughout a city. Two macroscopic fundamental diagrams are plotted, one for mixed-traffic in the absence of converted lanes, and the other for a scenario with the bus-only lane activated. Two corresponding maximum discharge rates are defined.

On the contrary, examinations of real-world vehicle trajectory data, like those from the NGSIM dataset, showed that the effective discharge rate may be discounted when merging or lane-changing behaviors are observed. In a time-space diagram, Figure 1 below presents two examples of vehicle trajectories from the NGSIM dataset, with X-axis being time and Y-axis being space dimension. The thick black lines represent two adjacent vehicles that are following each other in the same lane, while the thick purple line represents a vehicle that first merges into, and then leaves this lane. Following the terminology from (Zheng et al. 2013), these three vehicles are named ‘leader”, “follower”, and “lane-changer”, respectively. We use h_{bef} and h_{aft} to denote the time headway between the leader and follower, before and after the lane-changer’s appearance, respectively. It can be found that h_{bef} is initially relatively small. After the lane-changer shows up, the follower slows down to maintain a safe distance and, after the lane-changer leaves this lane, the follower accelerates again. However, the new time headway h_{aft} is significantly larger than h_{bef}, and may never be restored back to h_{bef}. As evidenced in both (a) and (b) of Figure 1, the blank white area between the leader and follower vehicles, that indicates the length of time that is not effectively utilized for traffic flow to go through, increased in both examples. As such, the effective discharge rate was discounted due to the impact of the lane-changer.

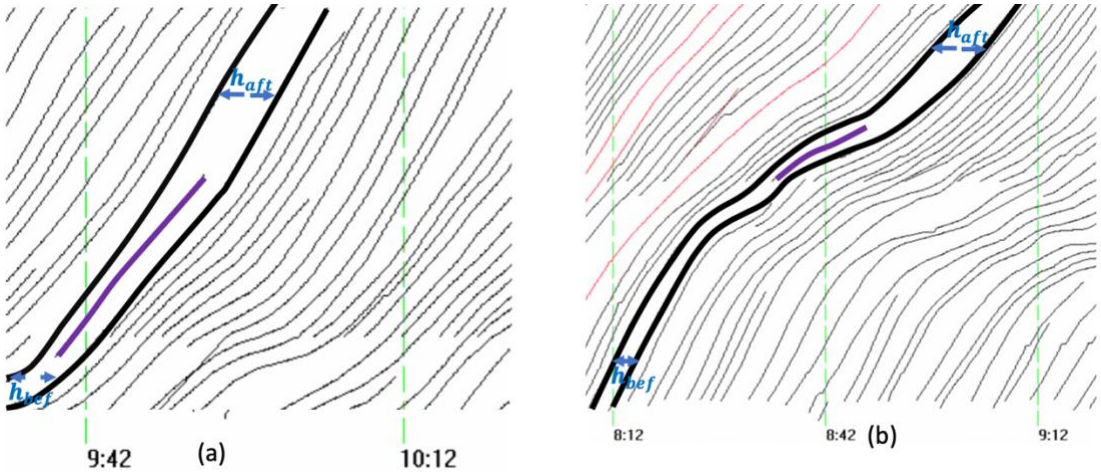


Figure 1. Illustration of discounted effective discharge rate with the NGSIM data and a time-space diagram.

To gain a better understanding of such a phenomenon, modeling efforts were made to analytically derive the reduced effective discharge rate of a roadway segment, considering the multi-stage lane-changing behavior and vehicular traffic kinematics near a side entrance. Here, multi-stage lane-changing behavior refers to the phenomenon that a general vehicle needs to wait at a side entrance for an acceptable gap, then merge into the bus-only lane, and quickly accelerate to leave for the general-purpose lane. We note that a bus-only lane is generally on the far-right side of the road, and such lane conversion may happen on both urban streets as well as on freeways. For example, Figure 2 (a) gives an example on arterial roadway with a bus-only lane, on 5th Street in downtown Los Angeles, whereas Figure 2 (b) gives another example on the Beijing 3rd Ring Road, a freeway with a speed limit of 80km/hr. In the arterial scenario, a stop sign is present at the side-street entrance, so an entering vehicle (i.e., lane-changer) has to stop and wait for an acceptable gap before merging onto the main road. The side-street entrance could be a shopping mall exit, a side street without signal control, or any other type of facility that generates traffic that needs to merge onto the main road. In the freeway scenario, a yield sign is present at the on-ramp merging location, so that an entering vehicle has to yield, and wait for an acceptable gap, before merging onto the main road. The entry speed is zero, for the arterial scenario, and a positive number for the freeway scenario.



(a) (b)

Figure 2. Illustration of bus-only lane conversion scheduling problem. (a) urban street example in Los Angeles, (b) freeway example in Beijing.

We are interested in the explicit derivation of effective discharge rate discount factors for this general roadway segment with a mixed traffic flow and, subsequently, the optimization of the bus-only lane conversion scheduling problem (i.e., should the right lane in Figure 2 be converted into a bus-only lane?). And, if yes, under what conditions, and what should the start and end times be? Note, a commonly-seen demand-oriented practice is to activate a bus-only lane only during peak hours. We argue that such a strategy, although easy to implement, does not guarantee an optimal traffic management outcome. Due to the nature of mixed traffic flow with the merging and lane-changing behaviors and heterogenous vehicular performance, the correct modeling of an effective discharge rate under different scenarios is critical for such quantitative decision making. However, to the best of our knowledge, the modeling efforts to derive an effective roadway discharge rate are rather limited in the literature, and are usually complicated by a lack of an explicit analytical expression.

This manuscript will fill this important research gap by deriving a simple, explicit, yet realistic effective discharge rate analytical expression. Three effective discharge rate discount factors are analytically derived, including one for bus-only lane, one for general-purpose lane in a scenario with bus-only lane activated, and the last one for mixed-traffic lane in a scenario with the bus-only lane deactivated. The formulation was based on multi-stage merging and lane-changing behavior, microscopic car-following behavior, shockwave propagation, and fundamental diagram. The classic fluid-based approximation method, by Newell, to characterize the queuing and dissipation process, is extended to a multi-state queuing analysis framework. The derived effective discharge rates are represented with mathematically-simple expressions in a closed-form, and correctly account for the effects of demand inputs (such as arrival rate and the ratios of vehicles of different types), vehicular performance (such as acceleration rate), and traffic flow characteristics (such as backward wave speed and jam density).

After correctly deriving an effective discharge rate, the scheduling of bus-only lane conversion is formulated as an optimization problem to determine the optimal start and end times of the bus-only lane, should it be activated at all. The numeric analysis showed that, when the NGSIM datasets from Lankershim Boulevard in Los Angles, and northbound I-80 in Emeryville, California, are used for testing arterial and freeway scenarios, respectively, the developed model reduced the effective discharge rate estimation error significantly. This suggested a satisfactory capability of capturing an effective discharge rate discount in a mixed traffic flow. In addition, the optimization model result showed that, when compared with the demand-oriented strategy of enabling a bus-only lane during peak hours, the proposed model reduces total traffic delay by 43.0% for the arterial scenario, and 48.2% for the freeway. Sensitivity analysis was also performed and showed that, when the arrival rate changes, or the ratio of vehicle types changes, the optimal bus-only lane scheduling time also varies. The results also indicated that, a bus-only lane conversion schedule is sensitive to input data and leads to a conclusion that determining a bus-only lane operation time that is based on time-varying inputs (instead of simply enabling it during peak hours), is more reasonable and can better reduce congestion.

This paper is organized as follows. Section 2 reviews relevant literature. Some preliminaries are provided in Section 3. Section 4 shows the derivations of three effective discharge rate discount factors, as mentioned above. Section 5 presents the bus-only lane conversion scheduling optimization model. The developed model is implemented and validated using the NGSIM dataset in Section 6, and the results are discussed. Section 7 concludes this paper.

2. Literature Review

Periodical reservation of some select street lanes for dedicated bus-only use is a commonly used strategy adopted by transit agencies, with the goal of increasing the transit's attractiveness and, ultimately, to reduce traffic congestion. However, a bus-only lane comes at the cost of reduced car-carrying capacity, which may lead to a long queue and excessive delay of general vehicles. Striking a balance between benefits to the bus and the damage to general traffic, thus becomes a significant challenge to address (Saade et al. 2018). Modeling efforts to optimally reallocate the road space between private cars and

transit modes can be found in Mesbah et al. (2011), Zheng and Geroliminis (2013), with the objective of minimizing system cost, which may include user travel time and/or operation cost. Innovative bus-only lane operation strategies, such as bus lanes with intermittent priority (BLIP) (Viegas and Lu 2004, Eichler and Daganzo 2006), pre-signal for bus-priority (Guler and Menendez 2014), and shared-lane strategy at a bottleneck (Guler and Cassidy 2012), are also reported in the literature.

Estimation of a bus-only lane effective discharge rate is important for its design and operation. While some researchers have studied bus-stop capacity, such as (Gu et al. 2011, Shen et al. 2019) and others, the estimation of an effective discharge rate for the bus-only lane, considering the impact of intrusion and the merging behavior of general vehicles entering roadway segment from a side entrance, is largely missing in the literature. This is important because, in reality, a bus-only lane is generally on the far-right side of a road and traffic is commonly observed merging from a side entrance that may severely affect the efficiency of the bus-only lane.

Modeling efforts to derive an effective roadway discharge rate were not often found in the literature, and were usually complicated and lacked explicit analytical expression. For example, Leclercq et al. (2011) proposed an analytical model that extended the Newell-Daganzo model by incorporating, endogenously, a capacity drop related to the merging process. As noted by the authors, there were two issues that may constrain the applicability of the developed model. First, for mathematical tractability, this model was not fully realistic because both incoming roads were assumed to be single-lanes and, second, an explicit formulation of capacity drop degree was impossible, as it had to be calculated by solving a predefined equation with a given set of parameters. This work was extended by Leclercq et al. (2016) by accounting for heterogeneous vehicle characteristics and the physical interactions between upstream waves and downstream voids, and by estimating the effective capacity at a freeway merger in a multilane context (Leclercq et al. 2016). However, an explicit analytical expression of capacity drop was still impossible, and a system of four equations and four unknowns were listed for computing the numerical solutions. Arnet et al. (2015) empirically analyzed differences in car capacity between a mixed-use scenario, without the influence of buses, and a dedicated lane scenario and a mixed-use scenario, with and without buses, with a focus on the ramp merging area. Yuan et al. (2017) investigated to what extent the acceleration spread and the reaction time could influence the queue discharge rate, as well as proposing a speed dependent reaction time mechanism for giving variable queue discharge rates. Tilg et al. (2018) studied how emerging automated vehicle technology can improve operations and increase the capacity of weaving sections, with a multiclass hybrid model. Laval and Daganzo (2006) focused on freeway sections, away from diverges, where the main incentive for drivers to change lanes was by increasing their speed. Mathematical models were developed to describe the mechanism of lane-changing vehicles, that created voids in traffic streams, and how these voids reduced flow. Chen and Ahn (2018) investigated how spatially distributed lane changes impact capacity-drop at an extended merge, diverge, and weave bottlenecks, and developed analytical models to capture the impacts of lane changes and numerical simulations to quantify capacity-drop. Other freeway capacity drop modeling efforts can be referred to (Menendez and Daganzo 2007, Laval and Leclercq 2008, Monamy et al. 2012, Parzani and Buisson 2012, Jin et al. 2015), among others. However, an explicit analytical expression of an effective discharge rate was not found in the literature, which greatly limits its applications in real-world problems.

Investigation of the bus-only lane conversion problem can also be found in the literature. Saade et al. (2018) took a simulation approach to analyze what can occur when select street lanes throughout a city are periodically reserved for buses. Findings revealed that activating conversions near the start of a rush could degrade travel, both by car and by bus, and that a balance was struck by activating lane conversions near the end of a rush, when vehicle accumulation in the city was at, or near, its maximum. The problem of when to activate and deviate the bus lane, however, was not addressed due to the lack of modeling and optimization efforts. In another study, Zheng et al. (2017) proposed a multimodal macroscopic fundamental diagram approach to capture multimodal traffic dynamics and travel costs by mode, and simulated in a Swiss city network to allocating a road space for bus lanes in cities. Work to optimize

transit road space priority at the network level can also be referred in Mesbah et al. (2011) and Mesbah et al. (2011). The focus was, however, on the identification of optimal roadway segments for implementation of a bus lane, rather than studying the bus lane activation and deviation time. Huang et al. (2020) proposed a hybrid bus operational scheme combining the bus lane reservation strategy and the stop-skipping control, in which a multi-stage stochastic programming model was developed to account for the uncertainty of passenger demand. Khoo and Ong (2015) proposed a bi-objective optimization model to optimize a bus lane schedule, taking into consideration bus and general vehicle performances. A hybrid nondominated sorting genetic algorithm, with a microscopic traffic simulation model. was developed to optimize a bus lane schedule. Research on bus-only lane conversions can also be referred to Safran et al. (2014), Chen (2015), Yu et al. (2015), and Zhao et al. (2019), among others. These research findings, although helpful in providing important insights into bus-only lane operation, did not go far enough to reveal the traffic flow characteristics of a bus-only lane. Learning from traffic flow fundamentals, when demand and discharge rates are known, the traffic delay can be analytically derived, which can be used as the objective for the optimal bus-only lane conversion problem. In the next three sections, we show how to analytically derive the effective discharge rate and, subsequently, how to formulate an optimization model to identify the bus-only lane activation and deactivation time.

3. Preliminaries

In this section, we briefly review the Newell simplified car-following model for the description of vehicle kinematic movement, to derive the associated effective discharge rate based on the fundamental diagram, and the fluid-based queue formation and dissipation process. Two main issues with the commonly-seen demand-oriented bus-only lane scheduling strategies are discussed. For convenient reference, the mathematical notations used in this paper are summarized in Appendix-A.

3.1 Newell Car-following behavior

A simplified car-following model was proposed by Newell (2002) and adopted in this manuscript to describe a vehicle's kinematic movements on a roadway segment. The simplified model states that the time-space trajectory of a following vehicle n is essentially the same as its leading vehicle $n-1$, except for a delay in space and in time. In the time-space diagram (shown in Figure 3 below), a leading vehicle $n-1$ initially drives at a constant speed v_1 and then changes to another constant speed v_2. Its actual trajectory is the dotted line but, for simplification, a piecewise linear approximation, represented by a solid line, is used. According to the simplified car-following model, the following vehicle n also travels at the same speed of v_1 at the beginning, and then changes to v_2. However, at the turning point, there is a temporal delay of τ_n which represents the driver n's necessary response time, as well as a spatial delay of d_n, which represents the distance needed to ensure safe driving.

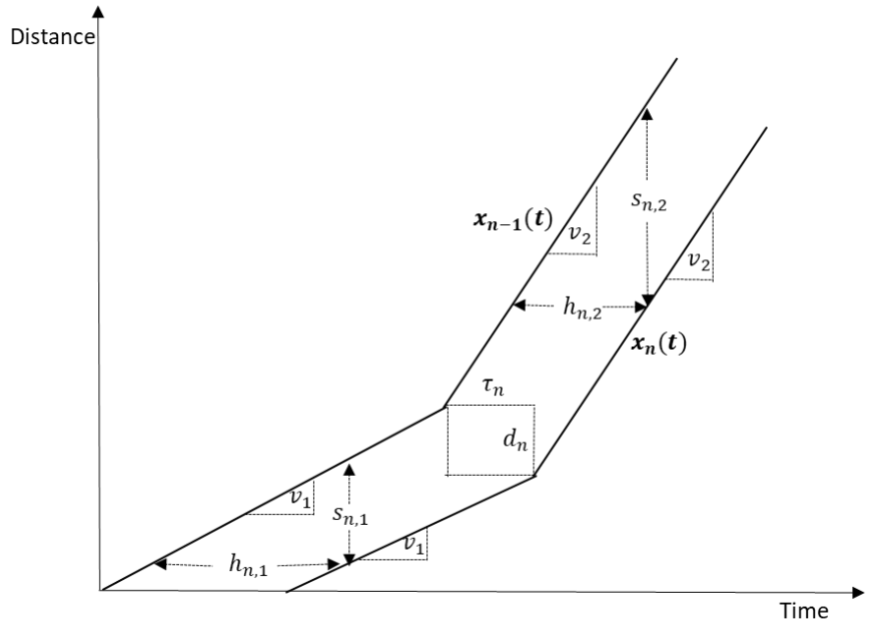


Figure 3. Time-space diagram of Newell's simplified car-following model.

The movement trajectory of the following vehicle n can be calculated by $x_n(t+\tau_n) = x_{n-1}(t) + d_n$. Based on the time-space diagram, the time headway of n's vehicle, before and after speed change, can be derived with $h_{n,1} = \tau_n + \frac{d_n}{v_1}$ and $h_{n,2} = \tau_n + \frac{d_n}{v_2}$, respectively. The space headway of n's vehicle, before and after the speed changes, can be computed by $s_{n,1} = d_n + \tau_n \cdot v_1$ and $s_{n,2} = d_n + \tau_n \cdot v_2$, respectively.

3.2 Discharge rate

The car-following model above suggests a triangular fundamental diagram. Per (Newell 2002), the q-k relationship can be represented as $q = \frac{1}{\bar{\tau}} - \frac{\bar{d}}{\bar{\tau}}k$, in which $\bar{\tau} = \frac{1}{n}\sum_{k=1}^{n}\tau_k$ and $\bar{d} = \frac{1}{n}\sum_{k=1}^{n} d_k$ .The fundamental diagram has a free-flow speed v_f, backward wave speed w, maximum discharge rate μ and its corresponding density k_m, and jam density k_j. Among the five variables, as long as three are known, the remaining two can be derived. Note in Figure 4 below, subscripts g, m, b are used to denote a general vehicle, mixed traffic, and bus, respectively. We also have the free-flow speed $v_{f,m}$ and $v_{f,b}$ for the other two modes, as well as w_m and w_b, (k_m^m, μ_m^m) and (k_b^m, μ_b^m), which are not noted in the figure for simplicity purposes.

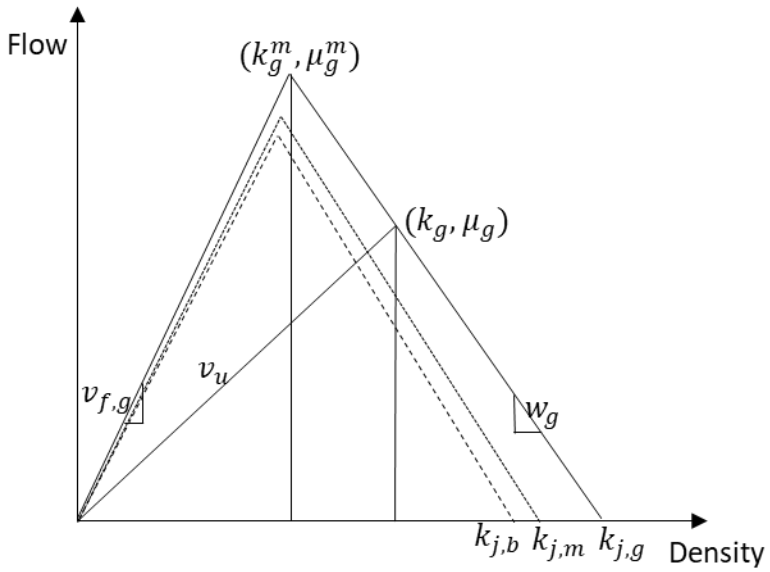


Figure 4. Triangular fundamental diagram.

Following the above fundamental diagram, when v_f, w, and k_j are known for a roadway segment, its maximum discharge rate μ^m (or μ_g^m for general vehicles) can be calculated as $\mu^m = \frac{k_j}{(\frac{1}{v_f}+\frac{1}{w})} = \frac{k_j \cdot v_f \cdot w}{v_f + w}$. It should be noted that μ^m is the maximum discharge rate under prevailing congested density. In a case, where there is a bottleneck (e.g., due to lane drop or other geometric reasons), the actual cruising speed may be lower than the free-flow speed. We used a new point (k_g, μ_g) in the q-k diagram in Figure 4 to represent the discharge rate due to the bottleneck impact on the general-purpose lane, with $\mu_g < \mu_g^m$, where μ_g^m is the effective discharge rate of general vehicles without the bottleneck effect. The cruising speed is denoted as v_u (v_g for the general-purpose lane, and v_b for the bus-only lane), with $v_u < v_f$, the new discharge rate becomes $\mu_g = \frac{k_{j,g} \cdot v_g \cdot w_g}{v_g + w_g}$. Since the formulations of μ^m (without a bottleneck) and μ_g (with a bottleneck) take the same form, for simplicity purposes, in the sections below we use μ to denote maximum discharge rate and v_u to denote cruising speed, and come to the generic formulation shown in Eq.(1) below.

$$\mu = \frac{k_j \cdot v_u \cdot w}{v_u + w} \tag{1}$$

3.3 Analyzing lane scheduling strategy from fluid-based approximation

Based on the classical work by Newell in using a fluid-based approximation method to characterize the queue formation and dissipation processes (Newell 1982), we considered two types of states: uncongested and congested. Under the congested state with homogeneous traffic, we considered a single value of μ. In Section 4, we further considered several types of congested states with a single type and with two types of vehicles, respectively.

We denoted the time-dependent arrival rate (i.e., the demand) as $\lambda(t)$. During off-peak hours, $\lambda(t) < \mu$ and no queue exists. When the demand increases and finally exceeds the μ, vehicles start to form a queue. The queue dissipates and traffic returns to normal, as demand reduces after peak hours. Figure 5 provides an illustration of the demand-supply relationship and queuing profile. In Figure 5(a), the demand (black

curve OABCE) exceeds the maximum discharge rate (purple horizontal line AD) during the peak hour between t_0 and t_2, and the queue formation is illustrated by the area highlighted by red lines (i.e., ABCA). The arrival rate reaches peak at time t_1. The longest queue is observed at time t_2 and starts to shrink, as shown by the green lines (i.e., area CDEC). The queue is fully discharged at time $\bar{t}$. The queuing profile is illustrated in Figure 5(b).

Based on the illustration in Figure 5, the length of the queue at time t can be derived by $Q(t) = \int_{t_0}^{t}(\lambda(\tau) - \mu)d\tau$. The total delay is thus the integral of queue length from t_0 to t_3, which can be calculated by Eq. (2) below.

$$W(t_3) = \int_{t_0}^{t_3} Q(t)d\tau = \int_{t_0}^{\bar{t}} Q(t)d\tau \tag{2}$$

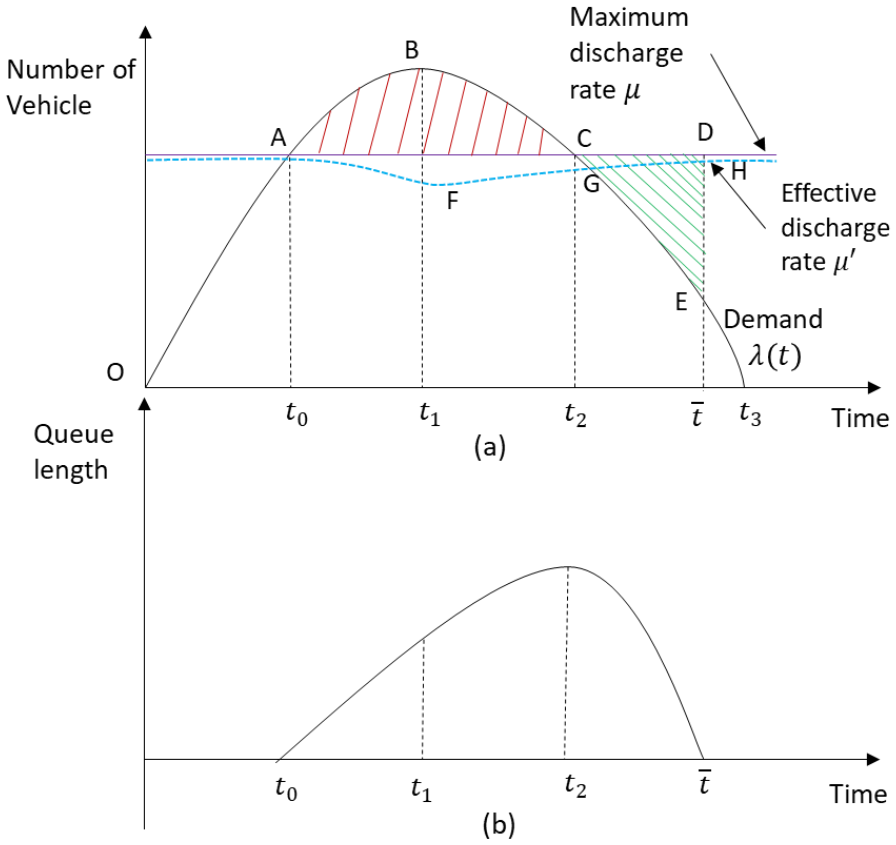


Figure 5. Queue formation and dissipation processes under a homogeneous traffic condition

As mentioned, a commonly-seen demand-oriented bus-only lane scheduling strategy is to activate a bus-only lane only during peak hours, i.e., at time $[t_0, t_2]$. We argue that such a strategy, although easy to implement, does not guarantee optimal traffic management outcomes, for two main reasons.

1. As illustrated with the NGSIM data example in Figure 1, the effective discharge rate may not be fixed at a value that is equal to the maximum discharge rate μ, instead, the effective discharge rate μ' (illustrated by the blue dash curve FGH in Figure 5(a)) might be lower than μ and fluctuate in response to different system inputs. As a result, the real queue length and vehicle delay will be different from those when the maximum discharge rate is used.

2. Although the peak hour ends at time t_2, according to the queuing profile in Figure 5(b), the queue length is the highest at t_2 and continues to cause vehicle delay. As such, the strategy of enabling a bus-only lane during peak hours $[t_0, t_2]$ does not make sense. On the other hand, continuing to activate the bus-only lane until time $\bar{t}$ might be an overkill, since the queue length sharply decreases and, as discussed previously, the bus-only lane comes with the cost of increased delays for general vehicles.

To deal with these issues, we show how to analytically derive μ', and quantitively determine a bus-only lane scheduling time in the next two sections.

4. Analytical Characterization of Effective Discharge Rate with a Multi-State Queuing Analysis Framework

In this section, we analytically derive the effective discharge rate for two scenarios. We start with a scenario where the bus lane is activated (scenario S_2), and derive the effective discharge rate μ' a for bus-only lane and a general-purpose lane, respectively. Then, we move on to a scenario where the bus-only lane is deactivated (scenario S_1), and derive μ' for the mixed traffic flow of both the bus and general vehicles. A simple roadway segment with two lanes and one side entrance is shown in Figure 6. It should be noted that the side entrance could be a side-street with a stop sign, as illustrated in Figure 2 (a), or an on-ramp merging onto the freeway, as illustrated in Figure 2 (b), with the key difference being the entry speed v_e. For the urban street case, $v_e = 0$ due to the presence of a stop sign, whereas, for the freeway case we have $v_e > 0$. We also note that the developed model can be iteratively applied to work with a road segment with multiple side-street entrances, or one that has three or more lanes.

4.1 Roadway with bus-only lane activated

First, we look at the scenario with a bus-only lane enabled. As shown in Figure 6 (a), the right-side lane is reserved for the buses, whereas all general vehicles drive in the left-side lane. An entering vehicle (i.e., the "lane-changer", or "merger") must wait for an acceptable gap before merging onto the main road, with an entry speed v_e. The flow rate of the bus in the bus-only lane is denoted as $\lambda_b(t)$, that of the general traffic vehicles on the general-purpose lane is $\lambda_g(t)$, and that of the entering vehicles that merge onto the main road from the side-street exit is $\lambda_e(t)$. If we denote the overall time-varying demand as $\lambda(t)$, and the time-varying proportion of the three types of vehicles as $\rho_b(t)$, $\rho_g(t)$, and $\rho_e(t)$, we have $\lambda_b(t) = \lambda(t) \cdot \rho_b(t)$, $\lambda_e(t) = \lambda(t) \cdot \rho_e(t)$, $\lambda_g(t) = \lambda(t) \cdot \rho_g(t)$, and $\rho_b(t) + \rho_e(t) + \rho_g(t) = 1$. Figure 6 (b) illustrates the triangular fundamental diagram, in which solid black lines represent the general-purpose lane and dashed red lines represent the bus-only lane. Similar to Figure 4, μ_g^m and μ_b^m are the maximum discharge rates under prevailing congested density, corresponding to the free-flow speeds $v_{f,g}$ and $v_{f,b}$. When a bottleneck is present, the actual cruising speeds become v_g and v_b, then the discharge rates due to the bottleneck impact are μ_g and μ_b, respectively. The backward wave speed on the general-purpose lane and the bus-only lane are w_g and w_b, respectively.

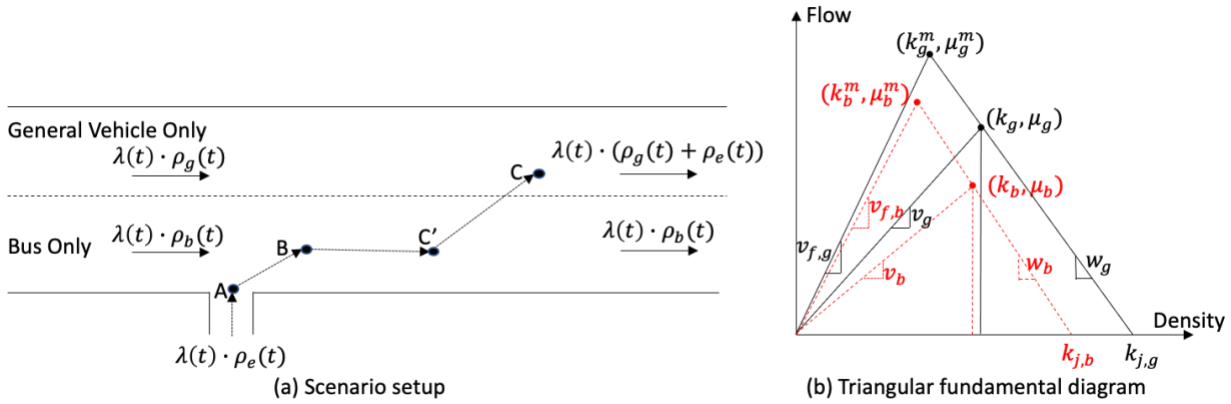


Figure 6. Scenario setup and triangular fundamental diagram with bus-only lane.

We break down the multi-stage movement process (as shown below) to describe the kinematic movements of the lane-changer in Figure 6:

1. Stage 1: Upon arrival at the side-street entrance at Point A at time t_A, the lane-changer will stop and wait for an acceptable gap in the traffic flow.
2. Stage 2: When a satisfactory gap is found (let's denote that as time t_B), he will merge into the bus-only lane, which corresponds to segment AB.
3. Stage 3: The lane-changer will accelerate to a higher speed (but no higher than free-flow v_f or cruising speed v_b) on the bus-only lane, which corresponds to segment BC' in Figure 6. To distinguish the cruising speeds on different lanes, we denote the cruising speed on the bus-only lane as v_b and that on the general-purpose lane as v_g.
4. Stage 4: Finally, the lane-changer will merge onto the general-purpose lane at Point C' at time $t_{C'}$ and accelerate to full cruising speed v_g (segment $C'C$) if he has not reached full cruising speed yet.

When the next lane-changer arrives at the side-street entrance location, the above steps 1~4 are repeated. If we denote the time of the next lane-changer's appearance as t_F (as shown in Figure 7), time $[t_A, t_F]$ becomes one episode with repeating patterns in the entire analysis period. As the time gap between t_A and t_F is the time headway of the lane-changer, we have $t_F - t_A = h_e = \frac{1}{\lambda(t)\cdot\rho_e(t)}$ in which $\lambda(t)$ is the arrival rate and $\rho_e(t)$ is the ratio of the lane-changer. It should be noted that corresponding to this multi-stage movement process, the queuing and dissipation process in one episode involves multiple states and multiple types of vehicles. In the two subsections below, we will extend Newell's classic fluid-based approximation method to a multi-state queuing analysis framework.

When analyzing this multi-stage process of the lane-changer, the following assumptions are made:

1. By the definition of a bus-only lane, the lane-changer cannot drive on the bus-only lane, other than using it for the purpose of transitioning onto a general-purpose lane. The lane-changer may switch to the general-purpose lane right after he merges onto the bus-only lane (i.e., at Point B), or may accelerate and then change lanes (i.e., switch while driving in before reaching Point C). This assumption requires drivers to observe traffic conditions, before enter the bus-only lane, and in the meantime, we also acknowledge that sometimes the vehicles in the general-purpose lane may yield to the lane changer, so that the lane changer does not have to stay in bus lane for long. However, driving on the bus-only lane at full cruising speed is prohibited, in any event.
2. The exact locations of a lane-changer merging onto the main road and then leaving the bus-only lane (i.e., Points B and C') are unknown. However, as analyzed above, we have $t_B \le t_{C'} \le t_C$. The empirical evidence in Daamen et al. (2010) shows that the merge location histogram in free-flow fits a lognormal distribution, whereas the merge location histogram in a congested state fits a

uniform distribution. Extensive tests have been performed in Leclercq et al. (2011) to demonstrate that a piece-wise uniform distribution leads to the same results, in terms of capacity drop, as a uniform distribution. Since this research studies the capacity drop, we mainly focus on the congested states, and, thus, adopt the assumption that the merge location follows a uniform distribution.

3. Vehicles accelerate with a constant acceleration rate a_g for general vehicles, and a_b for buses.
4. The traffic flow from a side exit is relatively low and, thus, the delay for a lane-changer can be omitted. In other words, we assume that h_e is high and multiple entering vehicles won't be observed arriving at the same time and forming a queue at the side-street entrance.

The multi-stage vehicle movements are further illustrated in Figure 7 below, in which Figure 7 (a) and (b) describe vehicle movements on the bus-only lane and on the general-purpose lane individually. X-axis is the time denoted by t_{point} and Y-axis is the distance denoted by x_{point}. The cruising speeds on the bus-only lane and on the general-purpose lane are v_b and v_g. We used red color to illustrate the trajectory of the lane-changer, and black solid lines for the vehicles in front of it and are not impacted by the lane-changer. The black dotted lines and the dashed lines represent vehicles affected by a moving bottleneck. The dotted lines illustrate the acceleration process, especially, and the dashed lines illustrate the trajectory after the vehicles' speed reaches the cruising speed. As discussed, an episode starts from Point A and ends at Point F, with a duration of $t_F - t_A = h_e = \frac{1}{\lambda(t)\cdot\rho_e(t)}$. Key action points of the lane-changer are A (arrival), B (merging into bus-only lane at speed v_e), C' (lane-changing into a general-purpose lane at a speed of $v_{C'}$) and C (fully accelerated to cruising speed v_g). Two shockwaves are generated on the bus-only lane, with one in time $[t_B, t_{C'}]$, in which a queue is formed after the lane-changer, and the other in $[t_{C'}, t_{E'}]$, in which the queue dissipates after the lane-changer switches to the general-purpose lane. Similarly, two shockwaves are generated on the general-purpose lane, one in time $[t_{C'}, t_C]$, in which a queue is formed after the lane-changer who is accelerating, and the other in time $[t_C, t_E]$, in which the queue dissipates after the lane-changer reaches cruising speed.

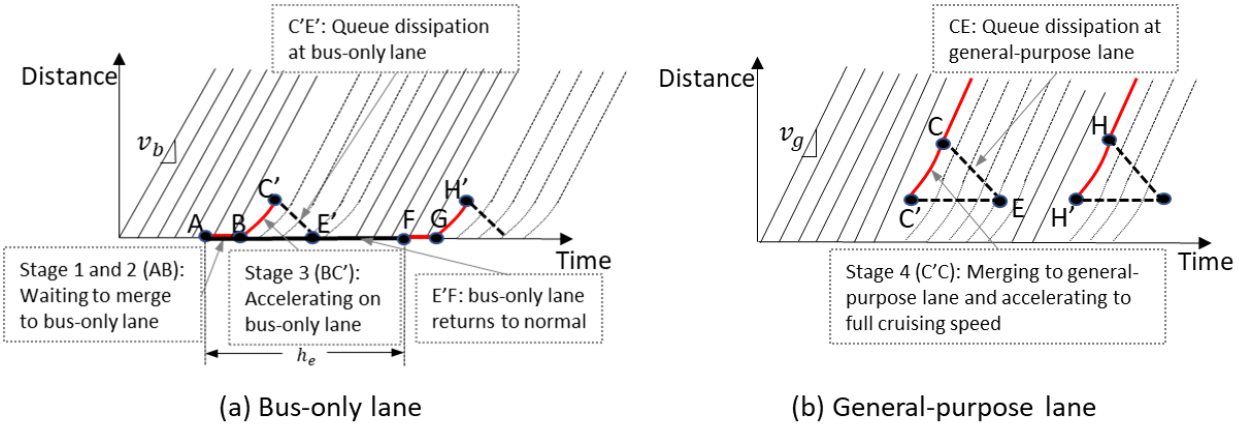


Figure 7. Multi-stage vehicle movement with a bus-only lane activated.

4.1.1 Effective discharge rate of the bus-only lane

First, let's derive the effective discharge rate for the bus-only lane (Figure 7(a)). As the lane-changer's position is constantly changing and it is the key element that influences the traffic dynamics, we firstly focus on analyzing the traffic flow rate that passes through shockwave boundary location C'. Note this reference point will moves upstream back to the merging location and, thus, all vehicles that are behind this moving bottleneck are counted, so we are not missing any vehicles.

At segment BC', the lane-changer accelerates on the bus-only lane at a speed that is below the cruising speed. In other words, the lane-changer becomes a moving bottleneck in the bus-only lane during time

$[t_B, t_{C\prime}]$. All buses are following the lane-changer, so the flow rate that passes this moving bottleneck C' location is 0. Assume that the lane-changer accelerates, with a constant value a_g, and we use $v_{c\prime}$ and v_c to denote the speed of the lane-changer at Points C' and C, respectively. As discussed earlier, the lane-changer cannot drive in a bus-only lane at full cruising speed, so we have the following equations for Point C', $v_e \le v_{c\prime} \le v_c = v_g$ for its speed, $t_B \le t_{c\prime} \le t_c = t_B + \frac{v_g - v_e}{a_g}$ for its time, and $v_e \le x_{C'} \le x_{C'} = \frac{{v_g}^2 - {v_e}^2}{2a_g}$ for its distance.

After the lane-changer switches to the general-purpose lane at Point C', the buses that were previously queued are being discharged at a speed of w_b and a density of $k_{j,b}$, at location C'. The effective discharge rate can be derived with Eq. (3), until all buses that have accumulated are discharged during time $[t_{C\prime}, t_{E\prime}]$.

$$\mu_{C\prime E\prime} = w_b \cdot k_{j,b} \tag{3}$$

As such, the effective discharge rate during time $[t_B, t_{E\prime}]$ can be derived as

$$\mu_{BE\prime} = \frac{(t_{C\prime} - t_B) \cdot \mu_{BC\prime} + (t_{E\prime} - t_{C\prime}) \cdot \mu_{C\prime E\prime}}{t_{E\prime} - t_B} = w_b \cdot k_{j,b} \cdot \frac{t_{E\prime} - t_{C\prime}}{t_{E\prime} - t_B} \tag{4}$$

Next, let's calculate the time differences between Points B, C', and E'. At Point B, the entry speed of the lane-changer is v_e, which has two scenarios. The first is an arterial scenario with an entry speed $v_e = 0$, and the second is a freeway scenario with an entry speed $v_e > 0$. At Point C' the speed is $v_{C'}$, and we know the vehicle accelerates with a constant acceleration rate, we have $t_{C'} - t_B = \frac{v_{c'} - v_e}{a_g}$. The spatial distance of Point C' can then be calculated as $x_{C'} = v_B \cdot (t_{C\prime} - t_B) + \frac{a_g}{2}(t_{C\prime} - t_B)^2 = \frac{{v_{c\prime}}^2 - {v_e}^2}{2a_g}$.

Between Points E' and C', the slow-moving vehicle queue is being discharged with a speed of w_b. Starting from location C', with spatial coordinate $x_{C'}$, the time difference between E' and C' is thus:

$$t_{E'} - t_{C'} = \frac{x_{C'}}{w_b} = \frac{{v_{c'}}^2 - {v_e}^2}{2a_g w_b} \tag{5}$$

So, the time differences between Points B and E' is given by

$$t_{E'} - t_B = (t_{E'} - t_{C'}) + (t_{C'} - t_B) = \frac{{v_{c'}}^2 - {v_e}^2}{2a_g w_b} + \frac{v_{c'}}{a_g} \tag{6}$$

Plug in Eqs. (5) and (6) into Eq. (4), the effective discharge rate between Points B and E' becomes

$$\begin{aligned} \mu_{BE'} &= w_b \cdot k_{j,b} \cdot \frac{t_{E'} - t_{C'}}{t_{E\prime} - t_B} = w_b \cdot k_{j,b} \cdot \frac{t_{E'} - t_{C'}}{(t_{E'} - t_{C'}) + (t_{C'} - t_B)} \\ &= \frac{w_b \cdot k_{j,b} \cdot (v_{C'} + v_e)}{v_{C'} + v_e + 2w_b} \end{aligned} \tag{7}$$

Note: between the lane-changer's arrival time t_A and the time a second lane-changer arrives at t_F, we also need to consider segments AB and $E'F$. For segment AB, the lane-changer is waiting for an acceptable gap, prior to merging onto the main road, which does not affect the traffic in the bus-only lane. For segment $E'F$, the lane-changer has left the bus-only lane, which does not affect the traffic in the bus-only lane either. In other words, during times t_{AB} and $t_{E'F}$ the traffic condition in the bus-only lane is normal and the discharge rate is μ_b. As such, Eq.(1) can be adjusted to get μ_b.

$$\mu_b = \frac{k_{j,b} \cdot v_b \cdot w_b}{v_b + w_b} \tag{8}$$

As segments AB, BC', $C'E'$, and $E'F$ constitute a time headway of the lane-changer h_e, we have

$$t_{AB} + t_{E'F} = h_e - t_{BE'} = h_e - \frac{{v_{c'}}^2 - {v_e}^2}{2a_g w_b} - \frac{v_{c'} - v_e}{a_g} \tag{9}$$

Thus, combining segments AB, BC', $C'E'$, and $E'F$, the adjusted effective discharge rate in the bus-only lane can be calculated via Eq. (10). For detailed derivation, please refer to Appendix-B.

$$\mu_b' = \frac{\mu_{BE'} \cdot t_{BE'} + \mu_b(t_{AB} + t_{E'F})}{h_e} = \mu_b \cdot \left(1 - \lambda(t) \cdot \rho_e \cdot \left(\frac{v_{c'} - v_e}{a_g} - \frac{{v_{c'}}^2 - {v_e}^2}{2a_g \cdot v_b}\right)\right) \tag{10}$$

Eq. (10) has a variable $v_{c'}$ which does not directly lead to an explicit effective discharge rate discount factor. In the arterial scenario, the entry speed $v_e = 0$. As the distribution of Point C' is assumed to be uniform, as discussed above, $v_{C'}$ is uniformly distributed between 0 and v_b. So we have $E(v_{c'}) = \frac{1}{v_b}\int_{v_e}^{v_b} v dv = \frac{v_b}{2}$, and $E({v_{c'}}^2) = \frac{1}{v_b}\int_0^{v_b} v^2 dv = \frac{{v_b}^2}{3}$. For the freeway scenario, the entry speed $v_e > 0$. As the distributions of Point B and Point C' are assumed to be uniform (as discussed above), v_e is uniformly distributed between 0 and v_b, and $v_{c'}$ are uniformly distributed between v_e and v_b. So we have $E(v_e) = \frac{1}{v_b}\int_0^{v_b} v dv = \frac{v_b}{2}$, $E({v_e}^2) = \frac{1}{v_b}\int_0^{v_b} v^2 dv = \frac{{v_b}^2}{3}$, $E(v_{c'}) = \frac{1}{v_b - v_e}\int_{v_e}^{v_b} v dv = \frac{3v_b}{4}$, $E({v_{c'}}^2) = \frac{1}{v_b - v_e}\int_{v_e}^{v_b} v^2 dv = \frac{11{v_b}^2}{18}$, and, thus, the expectation of an effective discharge rate for the bus-only lane can be derived via Eq. (11) below.

$$E[\mu_b'(t)] = \begin{cases} \mu_b \cdot \left(1 - \dfrac{\lambda(t) \cdot \rho_e(t) \cdot v_b}{3a_g}\right), \text{Arterial scenario} \\ \mu_b \cdot \left(1 - \dfrac{\lambda(t) \cdot \rho_e(t) \cdot v_b}{9a_g}\right), \text{Freeway scenario} \end{cases} \tag{11}$$

In other words, the discount factor of the effective discharge rate for the bus-only lane is $\frac{\lambda(t)\cdot\rho_e(t)\cdot v_b}{3a_g}$ for the arterial scenario and $\frac{\lambda(t)\cdot\rho_e(t)\cdot v_b}{9a_g}$ for the freeway scenario. Such formulations show that the capacity discount is mainly affected by arrival rate $\lambda(t)$, and entering vehicle ratio $\rho_e(t)$, as well as the bus cruising speed v_b, and that the impact of a merging vehicle is higher on arterials than on freeways. This is understandable, as when $\lambda(t)$ increases, traffic gets more congested and the effective discharge rate drops more significantly. When $\rho_e(t)$ increases, more entering vehicles are present and add a heavier discount to the effective discharge rate. In addition, the bus-only lane with a higher cruising speed v_b is impacted more by the merging vehicles, as the buses need to go through more drastic deceleration and acceleration processes.

4.1.2 Effective discharge rate of general-purpose lane

Next, we derive the effective discharge rate for the general-purpose lane, by examining the process of the lane-changer switching from the bus-only lane to the general-purpose lane, and the subsequent acceleration. Again, we use Figure 7 (b) to illustrate this process, and we focus on analyzing the traffic flow rate that passes through location C. The lane-changer switches from the bus-only lane to the general-purpose lane at a speed of $v_{C'}$, and accelerates to full cruising speed v_g at Point C. In other words, the lane-changer is a moving bottleneck in the general-purpose lane, and the general vehicles that are driving on the general-purpose lane start to form a queue during time $[t_{C'}, t_C]$. The moving bottleneck disappears and the queue starts to dissipate, once the lane-changer reaches cruising speed at Point C. A backward wave starts at Point C and the intersection with X-axis at Point E.

First, let's look at time $[t_{C'}, t_E]$, which can be decomposed into two segments, $[t_{C'}, t_C]$ and $[t_C, t_E]$. Similar to the process in the bus-only lane, during time $[t_{C'}, t_C]$ the vehicles in the general-purpose lane are following the moving-bottleneck and, thus, the flow rate that passes the moving bottleneck is 0. On the other hand, during time $[t_C, t_E]$, the accumulated vehicles are discharged with a discharge rate of μ_{CE}

$$\mu_{CE} = w_g \cdot k_{j,g} \tag{12}$$

As such, during time $[t_{C'}, t_E]$ the effective discharge rate can be derived as

$$\mu_{C'E} = \frac{(t_C - t_{C'}) \cdot \mu_{C'C} + (t_E - t_C) \cdot \mu_{CE}}{t_E - t_{C'}} = w_g \cdot k_{j,g} \cdot \frac{t_E - t_C}{t_E - t_{C'}} \tag{13}$$

Next, let's calculate the time differences between Points C', C, and E. As the lane-changer accelerates with a constant acceleration rate a_g between Points C' and C, we have $t_C - t_{C'} = \frac{v_g - v_{C'}}{a_g}$. The distance of Points C is $x_{CC'} = v_{C'} \cdot (t_C - t_{C'}) + \frac{a_g}{2} \cdot (t_C - t_{C'})^2 = \frac{{v_g}^2 - {v_{C'}}^2}{2a_g}$.

A backward wave starts from Points C and reaches Point E with a speed of w_g, so

$$t_E - t_C = \frac{x_{CC'}}{w_g} = \frac{{v_g}^2 - {v_{C'}}^2}{2a_g w_g} \tag{14}$$

As such, we have

$$t_E - t_{C'} = (t_E - t_C) + (t_C - t_{C'}) = \frac{v_g - v_{C'}}{2a_g w_g} \cdot (v_g + v_{C'} + 2w_g) \tag{15}$$

Plug Eq. (15) into Eq. (13), we can then infer the effective discharge rate $\mu_{C'E}$ between time C' and E as

$$\mu_{C'E} = w_g \cdot k_{j,g} \cdot \frac{t_E - t_C}{t_E - t_{C'}} = w_g \cdot k_{j,g} \cdot \frac{v_g + v_{C'}}{v_g + v_{C'} + 2w_g} \tag{16}$$

Note: In addition to time $[t_{C'}, t_E]$, between the lane-changer's arrival time t_A and the time a second lane-changer arrives at t_F, we also need to consider segments AC' and EF. For segment AC', the lane-changer does not yet travel in the general-purpose lane, so the traffic operation there is not affected. For segment EF, the lane-changer is driving at cruising speed, which does not affect the traffic operation either. In other words, during time $t_{AC'}$ and t_{EF}, the traffic condition in the general-purpose lane is normal and the discharge rate is μ_g. Eq.(1) can be adjusted to get μ_g for the general vehicles.

$$\mu_g = \frac{k_{j,g} \cdot v_g \cdot w_g}{v_g + w_g} \tag{17}$$

As segments AC', $C'C$, CE, and EF constitute a time headway of the lane-changer h_e, we have $t_{AC'} + t_{EF} = h_e - t_{C'E} = h_e - \frac{v_g - v_{C'}}{2a_g w_g} \cdot (v_g + v_{C'} + 2w_g)$. Thus, combining segments AC', $C'C$, CE, and EF, the adjusted effective discharge rate in the general-purpose lane can be calculated via Eq. (18). For a detailed derivation, please refer to Appendix-B.

$$\mu_g{}' = \frac{\mu_{C'E} \cdot t_{C'E} + \mu_g \cdot (t_{AC'} + t_{EF})}{h_e} = \mu_g \cdot \left(1 - \frac{\left(v_g - v_{C'}\right)^2 \cdot \lambda(t) \cdot \rho_e(t)}{2a_g \cdot v_g}\right) \tag{18}$$

Again, for the arterial scenario, the entry speed $v_e = 0$. As the distribution of Points C' is assumed to be uniform (as discussed above), $v_{C'}$ is uniformly distributed between 0 and v_b. So we have $E(v_{C'}) = \frac{v_b}{2}$, $E({v_{C'}}^2) = \frac{{v_b}^2}{3}$. For the freeway scenario, the entry speed $v_e > 0$. As the distribution of Points C' is assumed to be uniform (as discussed above), $v_{C'}$ is uniformly distributed between v_e and v_b. So we have $E(v_{C'}) = \frac{3v_b}{4}$, $E({v_{C'}}^2) = \frac{11{v_b}^2}{18}$, $E\left(v_g - v_{C'}\right)^2 = \mathrm{E}\left({v_g}^2 - 2v_g \cdot v_{C'} + {v_{C'}}^2\right) = {v_g}^2 - \frac{3}{2} v_g v_b + \frac{11}{18} {v_b}^2$.

As such, the expectation of an effective discharge rate for the general-purpose lane can be derived via Eq. (19) below.

$$E[\mu_g^{'}(t)] = \begin{cases} \mu_g \cdot \left[1 - \frac{\lambda(t) \cdot \rho_e(t)}{2a_g} \cdot \left(1 - \frac{v_b}{v_g} + \frac{{v_b}^2}{3{v_g}^2}\right)\right], & \text{arterial scenario} \\ \mu_g \cdot \left[1 - \frac{\lambda(t) \cdot \rho_e(t)}{2a_g} \cdot \left(1 - \frac{3v_b}{2v_g} + \frac{11{v_b}^2}{18{v_g}^2}\right)\right], & \text{freeway scenario} \end{cases} \quad (19)$$

In other words, the discount factor of the maximum discharge rate for the general-purpose lane is $\frac{\lambda(t)\cdot\rho_e(t)}{2a_g} \cdot \left(1 - \frac{v_b}{v_g} + \frac{{v_b}^2}{3{v_g}^2}\right)$ for the arterial scenario and $\frac{\lambda(t)\cdot\rho_e(t)}{2a_g} \cdot \left[1 - \frac{3v_b}{2v_g} + \frac{11{v_b}^2}{18{v_g}^2}\right]$ for the freeway scenario. Again, such formulations show that the effective discharge rate drops more significantly when $\lambda(t)$ or $\rho_e(t)$ increases. Comparing $1 - \frac{v_b}{v_g} + \frac{{v_b}^2}{3{v_g}^2}$ and $1 - \frac{3v_b}{2v_g} + \frac{11{v_b}^2}{18{v_g}^2}$, since we know the cruising speed of the bus-only lane is lower than that of the general purpose lane, i.e. $\frac{v_b}{v_g} < 1$, it can be easily proven that the discount factor is higher on the arterial than on the freeway. In addition, it can also be proven that, when $\frac{v_b}{v_g}$ increases (i.e., when cruising speeds become closer between the bus-only lane and the general-purpose lane), the discount factor decreases. In other words, a more homogenous traffic flow can help avoid drastic discharge rate drops. To prove this, we can let $x = \frac{v_b}{v_g}$, and easily show $\frac{\partial(-x+\frac{x^2}{3})}{\partial x} < 0$ and $\frac{\partial(-\frac{3}{2}x+\frac{11x^2}{18})}{\partial x} < 0$, as $x = \frac{v_b}{v_g} < 1$.

4.2 Roadway with bus-only lane deactivated

In this section, we work on a case with the bus-only lane deactivated (as illustrated in Figure 8 (a)). based on the fundamental diagram with moving coordinates. Again, we have an arterial scenario with zero entry speed and a freeway scenario with non-zero entry speed. As both lanes serve both buses and general vehicles, the total in-flow rate is now $\lambda(t) \cdot (\rho_b(t) + \rho_g(t))$ for both lanes on the main road, and the in-flow rate of the vehicles entering from the side-street exit is $\lambda(t) \cdot \rho_e(t)$. The lane flow distribution is represented by α, which means the in-flow rate on the left lane is $\alpha \cdot \lambda(t) \cdot (\rho_b(t) + \rho_g(t))$ and that on the right lane is $(1-\alpha) \cdot \lambda(t) \cdot (\rho_b(t) + \rho_g(t))$. Similar to what is discussed above, we have $\lambda_b(t) = \lambda(t) \cdot \rho_b(t)$, $\lambda_e(t) = \lambda(t) \cdot \rho_e(t)$, $\lambda_g(t) = \lambda(t) \cdot \rho_g(t)$, and note $\rho_b(t) + \rho_e(t) + \rho_g(t) = 1$. Again, Figure 8 (b) illustrates the triangular fundamental diagram, in which μ_m^m is the maximum discharge rate under prevailing congested density, corresponding to the free- flow speed $v_{f,m}$. The actual cruisng speed is v_u, and the discharge rate, due to the bottleneck impact, is μ_m with a backward wave speed w_m.

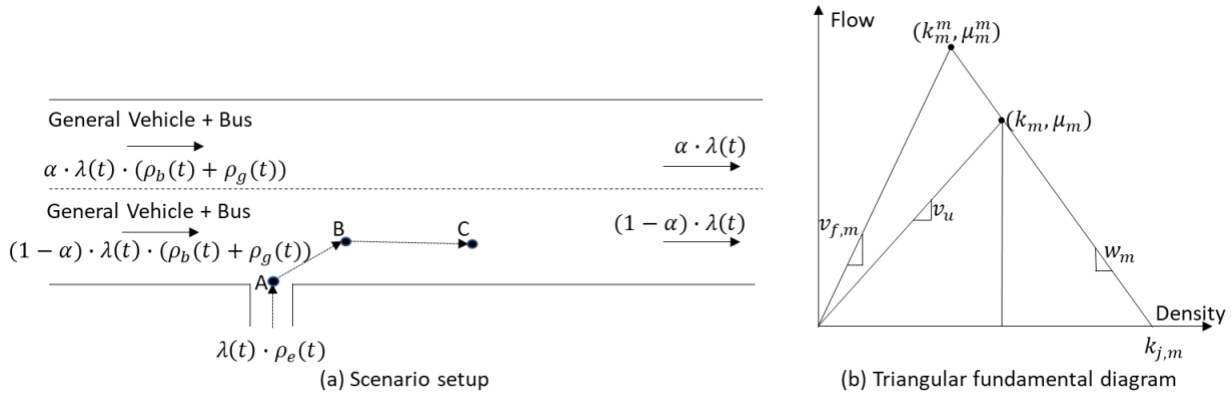


Figure 8. Scenario setup and triangular fundamental diagram without a bus-only lane.

As illustrated in Figure 8 (a), when arriving at a side-street entrance at Point A, the lane-changer will wait for an acceptable gap in the traffic flow. Once an acceptable gap is found, the lane-changer will merge onto the right lane of the main road (segment AB) at a speed of v_e, and then accelerate to cruising speed v_u, with a constant acceleration rate a_g (segment BC). Again, in the freeway scenario, v_e is uniformly distributed between 0 and v_u, so we have $E(v_e) = \frac{v_u}{2}$, $E({v_e}^2) = \frac{{v_u}^2}{3}$. The speed profile of the lane-changer can be represented by Eq. (20).

$$v^*(t) = \begin{cases} v_e & \text{if } t \le t_B \\ v_e + a_g \cdot (t - t_B) & \text{if } t_B < t < t_B + \dfrac{v_u - v_e}{a_g} \\ v_u & \text{otherwise} \end{cases} \tag{20}$$

Before the lane-changer's speed reaches v_u, it can be considered as a moving bottleneck in a two-lane roadway segment, so that the vehicles behind the lane-changer have to switch to the left lane to bypass the bottleneck. After they pass the moving bottleneck, the vehicles on the main road will mix. Considering the traffic flow distribution between lanes (Duret et al. 2012), the flow downstream becomes $\alpha \cdot \lambda(t)$ for the left lane, and $(1 - \alpha) \cdot \lambda(t)$ for the right lane.

Let's analytically derive the effective discharge rate with this moving bottleneck. Compared with a slowly moving truck, that travels at a constant speed, as discussed by Newell (1998), the unique characteristics of this moving bottleneck are: 1) the lane-changer is constantly accelerating in this process, so the speed of this moving bottleneck is time varying; 2) the arrival rates on the main roads $\lambda_b(t)$ and $\lambda_g(t)$ being studied are both time-dependent. These characteristics make the problem more complex to solve.

We start by placing a "moving observer" on the main road, who is travelling at the same velocity $v^*(t)$ as the lane-changer at any time t, at a location $x^*(t)$ relative to the lane-changer, which is derived in Eq. (21). We also place a "stationary observer" on the main road, whose location does not change.

$$x^*(t) = x(t) - v^*(t) \cdot t \tag{21}$$

As the moving observer moves at the same speed as the lane-changer, from his perspective, the two-lane roadway segment becomes a stationary section of road with only one lane for the other vehicles adjacent to the lane-changer. From his perspective, there is a lane reduction with a capacity drop. Some vehicles merge from two lanes to one lane, pass this bottleneck location, and then switch lanes to drive again on a two-lane roadway segment. Let's call the view from the moving observer's perspective "moving coordinates", and that from the stationary observer's perspective "stationary coordinates".

Assume that the flow and the density for the one-lane and the two-lane have a functional relationship in the stationary coordinates, i.e. $q = Q_1(k)$ and $q = Q_2(k)$, if we use $q^*(t)$ to denote the rate at which vehicles on the main road pass the moving observer, who is travelling at a speed of $v^*(t)$, then

$$q^*(t) = q - k \cdot v^*(t) \tag{22}$$

The density of the one-lane and the two-lane roadway segments is the same, no matter whose perspective (moving observer or stationary observer) is adopted. The relationship between $q^*(t)$ and k can be derived as

$$q^*(t) = Q_1(k) - k \cdot v^*(t) = {Q_1}^*(k), \qquad q^*(t) = Q_2(k) - k \cdot v^*(t) = {Q_2}^*(k) \tag{23}$$

in which ${Q_1}^*(k)$ and ${Q_2}^*(k)$ represent the q-k relationship on a one-lane and a two-lane roadway segment, as seen from the moving observer's perspective. The tricky part, in this q-k relationship, is that $v^*(t)$ is now time varying, indicating that, as time goes, the q-k relationship is also changing.

To make the discussion easier, we first present the 2-dimensional q-k fundamental diagram of a one-lane and a two-lane roadway, from both a moving observer's and a stationary observer's perspectives in Figure 9 (a). The moving observer's view is marked with red (i.e., OGI), and the stationary observer's view is marked with black (i.e., $OG'I'$). In this 2-dimensional diagram, at any given snapshot t, the moving bottleneck speed is fixed at $v^*(t)$. We start by noting that the triangle O-F-L is the fundamental diagram of the one-lane segment, and triangle O-G-I is the fundamental diagram of the two-lane segment in the moving coordinate. As discussed by Newell (1998), if we add a flow of $k \cdot v^*(t)$ to q^*, the resulting FD will be in the stationary coordinates. As such, the triangle O-F'-L' is the fundamental diagram of the one-lane segment and triangle O-G'-I' is the fundamental diagram of the two-lane segment in the stationary coordinates. We have the angle of $\angle G'OI' = arctan\, v_u$, angle of $\angle I'OI = arctan\, v^*(t)$, and the angle of $\angle G'I'O = arctan\, w$.

Before we go further, we hope to add a note to accurately describe the full picture that the 2-dimensional diagram needs to be extended to become 3-dimentional by adding an extra dimension of "moving bottleneck speed". As stated above, in this special problem, the moving bottleneck (i.e., the lane-changer) is in the process of accelerating from zero speed to full cruising speed and, thus, the 2-dimensional diagram in Figure 9 (a) can only capture the q-k relationship of one particular snapshot. If the full process is to be accurately described, a 3-dimensional diagram, like the one in Figure 9 (b), has to be used. By observing the new diagram, it can be found that, as the moving bottleneck's speed increases (as time goes), the flow rate from the stationary observer's view also increases. This is consistent with our general understanding that, as the moving observer is driving at a higher speed, the degree that traffic flow is blocked will decrease. We will come back to this 3-dimensional diagram in Figure 9 (b) when we derive an effective discharge rate but, for the sake of simplicity, we will focus on the 2-dimensional diagram in Figure 9 (a) first.

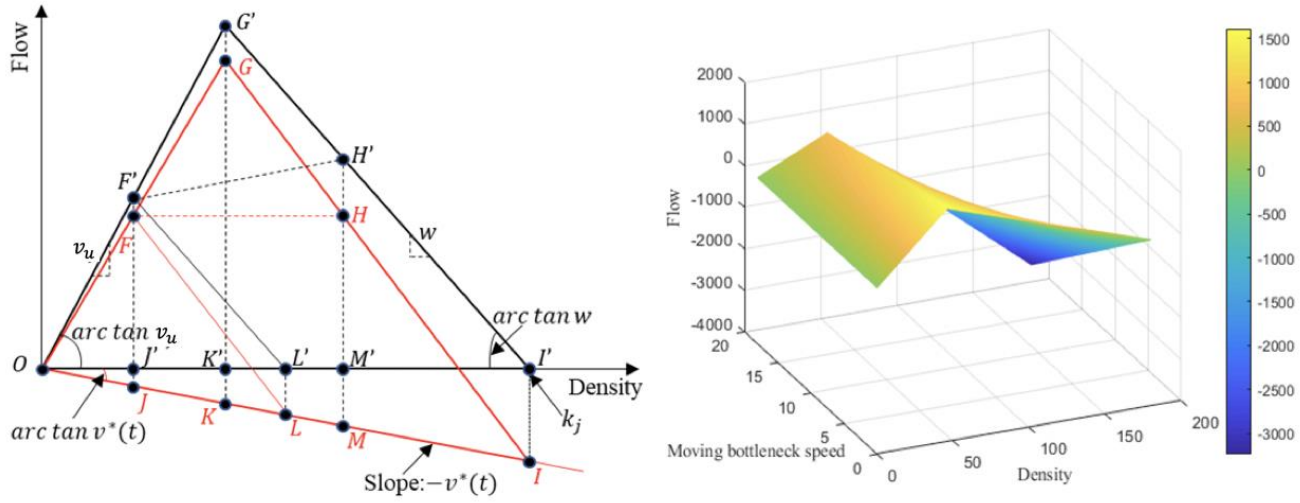


Figure 9. (a) 2-D diagram of the q-k relationship from a moving observer's view and a stationary observer's view; (b) 3-D diagram of the q-k relationship from a stationary observer's view.

We first calculate the adjusted flow rate $\lambda_{in}{}'(t)$ in the moving coordinate system. At any given time t, we have an incoming flow rate $\lambda_{in}(t)$ and the lane-changer's speed $v^*(t)$. To convert that into the adjusted flow rate $\lambda_{in}{}'(t)$ in the moving coordinate system, we back calculate the density on the two-lane roadway segment with $Q_2(k_2) = \lambda_{in}(t)$, or $k_2 = Q_2{}^{-1}(\lambda_{in}(t))$. Then, the adjusted flow rate $\lambda_{in}{}'(t)$ can be calculated as

$$\lambda_{in}{}'(t) = \lambda_{in}(t) - k_2 \cdot v^*(t) = \lambda_{in}(t) - Q_2{}^{-1}(\lambda_{in}(t)) \cdot v^*(t) \quad (24)$$

For example, in the stationary coordinate system, $F'J'$ is the maximum discharge rate of the one-lane segment, in which $F'J'$ is perpendicular to OI', and OJ' is its corresponding density. Following this with Eq.(24), to calculate the maximum discharge rate of the one-lane segment in the moving coordinate system, we need to reduce $F'J'$ by $OJ' \cdot v^*(t)$ which is JJ' and, thus, becomes FJ'. In other words, FJ' is the maximum discharge rate of the one-lane segment in the moving coordinate system.

In the moving coordinate system, if ${\lambda_{in}}'(t) < FJ'$, all vehicles can pass through the moving bottleneck at full cruising speed v_u. Otherwise, the adjusted arrival rate ${\lambda_{in}}'(t)$ becomes higher than the maximum discharge rate and, thus, the lane-changer becomes a moving bottleneck, and the roadway segment is subject to a maximum discharge rate with the value of FJ'. Subsequently, the traffic state of the downstream bottleneck location is represented by Point F, while that of the bottleneck upstream is represented by Point H, which has the same outflow rate as Point F (i.e., $y_F = y_H$,), but with a higher density due to the queue. When we convert the moving coordinates back to the stationary coordinates, Points F and H become Points F' and H', after adding a flow of $k \cdot v^*(t)$, i.e. $y_{F'} = y_F + k_F v^*(t)$ and $y_{H'} = y_H + k_H v^*(t)$. Since the density at Point H is higher than that at Point F (i.e., $k_H > k_F$), the flow rate at Point H' is also higher than that at Point F', i.e. $y_{H'} > y_{F'}$. Thus, it is Point H' that determines the maximum discharge rate, instead of Point F' in the stationary coordinate system.

Next, we derive the length of $H'M'$, which is the maximum discharge rate of Point H' in the stationary coordinate system. First, let's look at the triangle of F-O-L, which is the fundamental diagram of a one-lane roadway. As $F'J'$ is the maximum discharge rate of the one-lane segment, we have $F'J' = \frac{\frac{k_{j,m}}{2}}{(\frac{1}{v_u}+\frac{1}{w_m})} = \frac{k_{j,m} \cdot v_u \cdot w_m}{2(v_u+w_m)}$.

As the angle of $\angle G'OI' = arctan\, v_u$, OJ' can be calculated as $OJ' = \frac{F'J'}{v_u} = \frac{k_{j,m} \cdot w_m}{2(v_u+w_m)}$. As the angle of $\angle G'I'O = arctan\, w$, $J'L'$ can be calculated as $J'L' = \frac{F'J'}{w_m} = \frac{k_{j,m} \cdot v_u}{2(v_u+w_m)}$. As the angle of $\angle I'OI = arctan\, v^*(t)$, JJ' can be calculated as $JJ' = OJ' \cdot v^*(t) = \frac{k_{j,m} \cdot w_m \cdot v^*(t)}{2(v_u+w_m)}$. Then the maximum discharge rate of the moving bottleneck in the moving coordinates, with the value of FJ', can be calculated as

$$FJ' = F'J' - JJ' = \frac{k_{j,m} \cdot v_u \cdot w_m}{2(v_u + w_m)} - \frac{k_{j,m} \cdot w_m \cdot v^*(t)}{2(v_u + w_m)} = \frac{k_{j,m} \cdot w_m \cdot (v_u - v^*(t))}{2(v_u + w_m)} \tag{25}$$

But, what we really need to calculate is the maximum discharge rate of Point H' in the stationary coordinates, i.e., the length of $H'M'$. In the moving coordinates, we have the same flow at Points F and H, which means that $HM' = FJ' = \frac{k_{j,m} \cdot w_m \cdot (v_u - v^*(t))}{2(v_u+w_m)}$. As $G'K'$ is the maximum discharge rate of the two-lane segment, we have $G'K' = \mu_m = \frac{k_{j,m}}{(\frac{1}{v_u}+\frac{1}{w_m})} = \frac{k_{j,m} \cdot v_u \cdot w_m}{v_u+w_m}$, and as the angle of $\angle G'I'O = arctan\, w$, $K'I'$ can be calculated as $K'I' = \frac{G'K'}{w_m} = \frac{k_{j,m} \cdot v_u}{v_u+w_m}$.

Suppose the density at Point H' is k, we have $w_m = \frac{H'M'}{M'I'} = \frac{HM' + kv^*(t)}{k_{j,m} - k} = \frac{\frac{k_{j,m} \cdot w_m \cdot (v_u - v^*(t))}{2(v_u+w_m)} + kv^*(t)}{k_{j,m} - k}$ and, thus, the density at Point H' can be calculated as $k = \frac{k_{j,m} \cdot w_m}{2}(\frac{1}{v_u+w_m} + \frac{1}{v^*(t)+w_m})$.

The maximum discharge rate at Point H' in the stationary coordinates, i.e., the length of $H'M'$ can be calculated via Eq. (26). For detailed derivation, please refer to Appendix-B.

$$H'M' = HM' + kv^*(t) = \mu_m \cdot \frac{2v_u \cdot v^*(t) + v^*(t) \cdot w_m + w_m \cdot v_u}{2(v^*(t) + w_m) \cdot v_u} \tag{26}$$

Until this point, the maximum discharge rate of the moving bottleneck in the 2-dimensional stationary coordinate system, in Figure 9 (a), is determined. However, Eq. (26) has a time varying term $v^*(t)$ which does not directly lead to an explicit effective discharge rate discount factor. As discussed above, when the entire moving bottleneck is shown in the 3-dimensional diagram in Figure 9 (b), the q-k diagram is expanded by adding an extra dimension of "moving bottleneck speed", with the value increasing from 0 to v_f.

As discussed earlier, the speed profile of the lane-changer is represented by Eq. (20), and the time spent to accelerate from v_e to v_u is $\frac{v_u - v_e}{a_g}$. On the other hand, we know that the headway of the lane-changer traffic from the side-street entrance is $\frac{1}{\lambda(t)\cdot\rho_e(t)}$. In other words, the moving bottleneck exists between $[0,\frac{v_u - v_e}{a_g}]$ with a linearly increasing moving bottleneck driving speed while, for $[\frac{v_u - v_e}{a_g}, \frac{1}{\lambda(t)*\rho_e(t)}]$, both lanes are open to traffic without interruption.

With this process described, next we can derive the effective discharge rate of this moving bottleneck. Between $[0,\frac{v_u - v_e}{a_g}]$, $v^*(t) = v_e + a_g t$, so $E_{\left[0,\frac{v_u - v_e}{a_g}\right]}[{\mu_m}'(t)]$ can be computed via Eq. (27). For detailed derivation, please refer to Appendix-B.

$$E_{\left[0,\frac{v_u - v_e}{a_g}\right]}[{\mu_m}'(t)] = \begin{cases} \frac{\mu_m}{v_u} \cdot \left[\frac{2v_u + w_m}{2} + \left(\frac{w_m}{2} + \frac{{w_m}^2}{2v_u}\right) \cdot \ln\left(\frac{w_m}{v_u + w_m}\right)\right], \text{arterial scenario} \\ \frac{\mu_m}{v_u} \cdot \left[\frac{2v_u + w_m}{2} + \frac{w_m(v_u + w_m)}{2(v_u - v_e)} \ln\left(\frac{v_e + w_m}{v_u + w_m}\right)\right], \text{freeway scenario} \end{cases} \tag{27}$$

For the time between $\frac{v_u - v_e}{a_g}$ and $\frac{1}{\lambda(t)*\rho_e(t)}$, the discharge rate is μ_m. Combined, the expectation of an effective discharge rate for this moving bottleneck in the moving coordinates can be computed by Eq.(28). For a detailed derivation, please refer to Appendix-B.

$$E[{\mu_m}'(t)] = \begin{cases} \mu_m \cdot \left(1 - \frac{w_m \cdot \lambda(t) \cdot \rho_e(t)}{2a_g} \cdot \left[\left(1 + \frac{w_m}{v_u}\right) \ln\left(1 + \frac{v_u}{w_m}\right) - 1\right]\right), \text{arterial} \\ \mu_m * \left\{1 - \frac{v_u \lambda(t) \rho_e(t)}{2a_g} \cdot \left[\frac{w_m}{v_u} \cdot \left(\frac{1}{2} + \frac{w_m}{v_u}\right) - \left(\frac{w_m}{v_u}\right)^2 \cdot \left(1 + \frac{w_m}{v_u}\right) \cdot \ln\left(1 + \frac{v_u}{w_m}\right)\right]\right\}, \text{freeway} \end{cases} \tag{28}$$

Let

$$\begin{gathered} \left[\left(1 + \frac{w_m}{v_u}\right) \ln\left(1 + \frac{v_u}{w_m}\right) - 1\right] = \varphi_1 \\ \left[\frac{w_m}{v_u} \cdot (\frac{1}{2} + \frac{w_m}{v_u}) - \left(\frac{w_m}{v_u}\right)^2 \cdot \left(1 + \frac{w_m}{v_u}\right) \cdot \ln\left(1 + \frac{v_u}{w_m}\right)\right] = \varphi_2 \end{gathered} \tag{29}$$

We will have

$$E[{\mu_m}'(t)] = \begin{cases} \mu_m \cdot \left(1 - \frac{w_m \cdot \lambda(t) \cdot \rho_e(t)}{2a_g} \cdot \varphi_1\right), \text{arterial scenario} \\ \mu_m \cdot \left(1 - \frac{v_u \cdot \lambda(t) \cdot \rho_e(t)}{2a_g} \cdot \varphi_2\right), \text{freeway scenario} \end{cases} \tag{30}$$

In other words, the discount factor of the effective discharge rate for a mixed traffic flow, without a bus-only lane is $\frac{w_m \cdot \lambda(t) \cdot \rho_e(t)}{2a_g} \cdot \varphi_1$ for the arterial scenario and $\frac{v_u \cdot \lambda(t) \cdot \rho_e(t)}{2a_g} \cdot \varphi_2$ for the freeway scenario, which is affected by several factors. As the arrival rate $\lambda(t)$ or proportion of entering vehicles $\rho_e(t)$ increases, more loss to the effective discharge rate will be observed. If an entering vehicle has a higher acceleration rate a_g, its impact on the effective discharge rate for the bus-only lane would be lower. The discount factor of an effective discharge rate is also related to the backward wave speed w_m and cruising speed v_u. However, the relationship is much more complicated without an apparent linear or monotonic relationship.

To summarize, the three derived effective discharge rates from Eqs.(11), (19), and (30) are listed in Table 1 below. A comparison of these three effective discharge rates reveals that all of them can be explicitly calculated in a closed-form, as long as the entering vehicle's demand information $\lambda(t)$ and $\rho_e(t)$, as well as the traffic flow property w_m, v_u, v_g, v_b, a_g are given. These are usually not difficult to obtain in reality.

Table 1 List of three derived effective discharge rates

Scenarios	Study lane	Effective Discharge Rate
With bus-only lane activated	Bus-only lane	$E[\mu_b'(t)] = \begin{cases} \mu_b \cdot \left(1 - \frac{\lambda(t) \cdot \rho_e(t) \cdot v_b}{3a_g}\right), \text{arterial scenario} \\ \mu_b \cdot \left(1 - \frac{\lambda(t) \cdot \rho_e(t) \cdot v_b}{9a_g}\right), \text{freeway scenario} \end{cases}$
	General-purpose lane	$E[\mu_g'(t)] = \begin{cases} \mu_g \cdot \left[1 - \frac{\lambda(t) \cdot \rho_e(t)}{2a_g} \cdot \left(1 - \frac{v_b}{v_g} + \frac{{v_b}^2}{3{v_g}^2}\right)\right], \text{arterial scenario} \\ \mu_g \cdot \left[1 - \frac{\lambda(t) \cdot \rho_e(t)}{2a_g} \cdot \left(1 - \frac{3v_b}{2v_g} + \frac{11{v_b}^2}{18{v_g}^2}\right)\right], \text{freeway scenari} \end{cases}$
No bus-only lane	Mixed traffic lane	$E[\mu_m'(t)] = \begin{cases} \mu_m \cdot \left(1 - \frac{w_m \cdot \lambda(t) \cdot \rho_e(t)}{2a_g} \cdot \varphi_1\right), \text{arterial scenario} \\ \mu_m \cdot \left(1 - \frac{v_u \cdot \lambda(t) \cdot \rho_e(t)}{2a_g} \cdot \varphi_2\right), \text{freeway scenario} \end{cases}$

5. Optimization of Bus-Only Lane Conversion

With the effective discharge rate discount factors derived for each scenario in Table 1, in this section we formulate the bus-only lane conversion as an optimization problem, with the goal of minimizing the total system delay. Note that the system delay could be the summation of either a vehicle delay or a passenger delay, which do not differ much in the modeling process. To refresh our memory, the bus-only lane conversion problem is summarized below.

We are interested in calculating the start time and end time of bus-only lane operation, so that the total traffic delay can be minimized. Note that, when the bus-only lane is activated, the roadway segment setup is as illustrated in Figure 6. The flow rate on the bus-only lane is $\lambda_b(t)$ and the effective discharge rate

discount factor is $\frac{\lambda(t)\cdot\rho_e(t)\cdot v_b}{3a_g}$ or $\frac{\lambda(t)\cdot\rho_e(t)\cdot v_b}{9a_g}$, while the flow rate of the general-purpose lane is $\lambda_g(t)$ but becomes $(\lambda_g(t)+\lambda_e(t))$ after the merging point, with an effective discharge rate discount factor of $\frac{\lambda(t)\cdot\rho_e(t)}{2a_g}\cdot\left(1-\frac{v_b}{v_g}+\frac{{v_b}^2}{3{v_g}^2}\right)$ or $\frac{\lambda(t)\cdot\rho_e(t)}{2a_g}\cdot\left(1-\frac{3v_b}{2v_g}+\frac{11{v_b}^2}{18{v_g}^2}\right)$. When the bus-only lane is deactivated, the flow rate is $\alpha\cdot\lambda(t)\cdot(\rho_b(t)+\rho_g(t))$ for the left lane, and $(1-\alpha)\cdot\lambda(t)\cdot(\rho_b(t)+\rho_g(t))$ for the right lane, before the merge point. After the merging point, the flow rate becomes $\alpha\cdot\lambda(t)$ for the left lane, and $(1-\alpha)\cdot\lambda(t)$ for the right lane. The effective discharge rate discount factor is $\frac{w_m\cdot\lambda(t)\cdot\rho_e(t)}{2a_g}\cdot\varphi_1$ or $\frac{v_u\cdot\lambda(t)\cdot\rho_e(t)}{2a_g}\cdot\varphi_2$, for the arterial and the freeway scenarios, respectively.

On a typical day, the traffic demand starts low, during which time the traffic is light and there is no need for a dedicated bus-only lane, so scenario S_1 is preferred. As peak-hour approaches, the demand increases quickly and we envision switching to scenario S_2 where a bus-only lane will be beneficial for some period of time. When the peak-hour is over, the traffic cools down and, again, we will switch back to scenario S_1 without a bus-only lane.

If we use $[t_s,t_e]$ to denote the analysis period, and $t_{s_2,s},t_{s_2,e}$ to denote the start and end time of scenario S_2, the objective function to minimize total delay can be derived as:

$$Min\ W=\int_{t_s}^{t_{s_2,s}}Q_{s_1,1}(t)dt+\int_{t_{s_2,s}}^{t_{s_2,e}}Q_{s_2}(t)dt+\int_{t_{s_2,e}}^{t_e}Q_{s_1,2}(t)dt \tag{31}$$

In which $Q_{s_1}(t)$ and $Q_{s_2}(t)$ represent the delay of scenarios 1 and 2 at a given time t. Since we start with S_1 and also end with S_1, there are two portions of delay for this scenario. Note that a commonly-seen demand-oriented practice is to activate a bus-only lane only during peak hours. For comparison purposes, we also list the delay of such operational strategy below:

$$W=\int_{t_s}^{t_0}Q_{s_1,1}(t)dt+\int_{t_0}^{t_2}Q_{s_2}(t)dt+\int_{t_2}^{t_e}Q_{s_1,2}(t)dt \tag{32}$$

with t_0,t_2 denoting the start and end time of the peak-hour, as illustrated in Figure 5, respectively. $Q_{s_1,1}(t)$, $Q_{s_2,b}(t)$, $Q_{s_2,g}(t)$ and $Q_{s_1,2}(t)$ can be calculated by Eqs. (33)~(37). Note Eq. (37) indicates that the queue length is continuous and has memory, i.e., when the bus-only lane is deactivated after peak-demand is over (the scenario switches from S_2 to S_1), the queue length at the end of scenario S_2 (calculated by Eq.(36)) is carried over to the beginning of S_1.

$$Q_{s_1,1}(t)=\int_{t_0}^{t}\left(\alpha\cdot\lambda(\tau)-E[{\mu_m}'(\tau)]\right)d\tau+\int_{t_0}^{t}\left((1-\alpha)\cdot\lambda(\tau)-E[{\mu_m}'(\tau)]\right)d\tau \tag{33}$$

$$Q_{s_2}(t)=Q_{s_2,b}(t)+Q_{s_2,g}(t) \tag{34}$$

$$Q_{s_2,b}(t)=\int_{t_{s_2,s}}^{t}\left(\lambda_b(\tau)-E[{\mu_b}'(\tau)]\right)d\tau \tag{35}$$

$$Q_{s_2,g}(t)=\int_{t_{s_2,s}}^{t}\left(\lambda_g(\tau)+\lambda_e(\tau)-E[{\mu_g}'(t)]\right)d\tau \tag{36}$$

$$\begin{aligned}Q_{s_1,2}(t)=&\,Q_{s_2}\left(t_{s_2,e}\right)\\&+\int_{t_{s_2,e}}^{t}\left(\alpha\cdot\lambda(\tau)-E[{\mu_m}'(\tau)]\right)d\tau+\int_{t_{s_2,e}}^{t}\left((1-\alpha)\cdot\lambda(\tau)-E[{\mu_m}'(\tau)]\right)d\tau\end{aligned} \tag{37}$$

Calculation of $E[\mu_b{}'(\tau)]$, $E[\mu_g{}'(t)]$ and $E[\mu_m{}'(\tau)]$ in Eqs.(11), (19) and (30) requires μ_b, μ_g and μ_m, which can be derived from (1) with some mode-specific modifications, which leads to $\mu_b = \frac{k_{j,b} \cdot v_b \cdot w_b}{v_b + w_b}$, $\mu_g = \frac{k_{j,g} \cdot v_g \cdot w_g}{v_g + w_g}$ and $\mu_m = \frac{k_{j,m} \cdot v_u \cdot w_m}{v_u + w_m}$.

A few constraints on the decision variables are also listed below.

$$\lambda_b(t) = \lambda(t) \cdot \rho_b(t), \lambda_e(t) = \lambda(t) \cdot \rho_e(t), \lambda_g(t) = \lambda(t) \cdot \rho_g(t) \tag{38}$$

$$\rho_b(t) + \rho_e(t) + \rho_g(t) = 1 \tag{39}$$

$$t_{s_2,s} \leq t_{s_2,e} \tag{40}$$

$$t_s \leq t_{s_2,s} \leq t_e \tag{41}$$

$$t_s \leq t_{s_2,e} \leq t_e \tag{42}$$

6. Numerical Analysis

6.1 Effective discharge rate validation

To validate the effective discharge rate discount factors that are derived in Section 4, the NGSIM dataset is used. The validation efforts focus on two datasets, with Lankershim Boulevard in Los Angles for the arterial scenario, and Northbound I-80 in Emeryville, California, for the freeway scenario, as shown in Figure 10. The study area on Lankershim Boulevard focuses on the two signal intersections, and we are mainly concerned about the impact of the merging vehicles from the side entrance (shown in Figure 10 (a)). The study area on I-80 is approximately 500 m in length with an on-ramp (shown in Figure 10 (b)). The length of the acceleration section is 43 m. As we are mainly concerned with the impact of the lane-changer on the effective discharge rate of the main road, lane 5 and lane 6 (which are adjacent to the on-ramp, i.e., lane 7) are chosen.

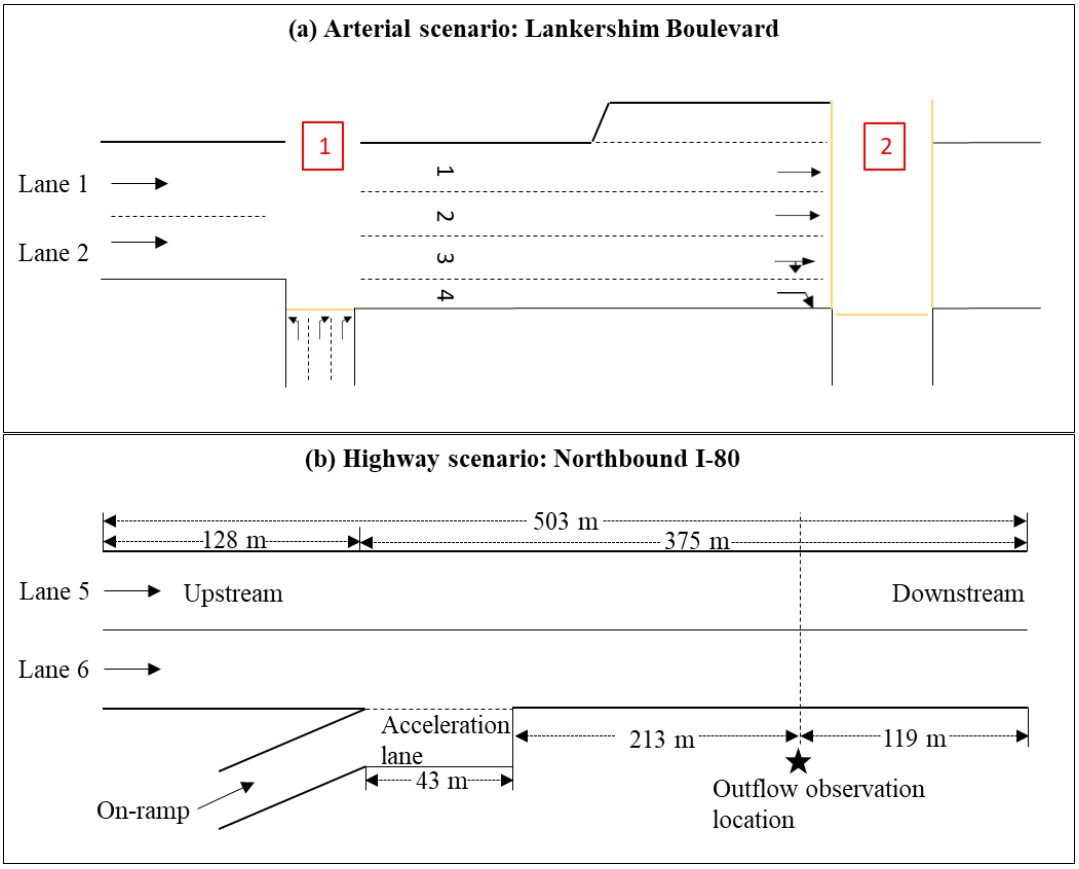


Figure 10. An illustration of the (a) Lankershim Boulevard and (b)Northbound I-80. study area.

An open-source software DTALite/NeXTA (Zhou and Taylor 2014) is used to import NGSIM vehicles trajectories and to extract the jam density, backward wave speed, and other needed characteristics. For the arterial scenario, the vehicle trajectories were collected between 08:30 am and 09:00 am on June 6, 2005, from Lankershim Boulevard. The cruising speed is set at 32 km/hr (or 20mph) and back wave speed is 13km/hr (or 8 mph) according to the trajectory from DTALite/NeXTA. The maximum discharge rate is $\mu_m = \frac{3k_{j,m} \cdot v_u \cdot w_m}{v_u + w_m} = 3{,}086$ veh/hr. The arrival rate can be observed as (704+74) *4=3,112 veh/hr and the ground truth discharge rate can be observed as 671 *4=2,684veh/hr. As the initial speed of merging vehicles on the arterial road is 0, the discount factor is $\frac{w_m \cdot \lambda(t) \cdot \rho_e(t)}{2a_g} \cdot \varphi_1$ based on Eq.(30). The expectation of an effective discharge rate can be calculated as $E(\mu_m') = 3{,}086 \times \left(1 - \frac{8 \times 74 \times 4}{2 \times 5 \times \frac{3600^2}{5280}} \times \left[\left(1 + \frac{8}{20}\right) \cdot \ln\left(1 + \frac{20}{8}\right) - 1\right]\right) = 2{,}861$ veh/hr.

Compared with the ground truth discharge rate, the absolute percentage error (APE) can be calculated as $APE_1 = \frac{|2861 - 2684|}{2684} = 6.6\%$. On the other hand, if we ignore the impact of lane-changers from the side entrance and directly use the maximum discharge rate μ_m, the APE would have become $APE_2 = \frac{|3086 - 2684|}{2684} = 15.0\%$. Comparison of APE_1 and APE_2 suggests that using the proposed effective discharge rate is more reasonable than directly using the maximum discharge rate, and the resulting APE value drops from 15.0% to 6.6%, a significant improvement.

Next, we focus validation of the freeway scenario. Figure 11 illustrates the vehicle trajectories in a time-space diagram for lane 5, lane 6, and the on-ramp (lane 7), respectively, between 04:00 p.m. and 04:15 p.m. on April 13, 2005. The area highlighted by black rectangles A to F highlight the shockwave propagation in the NGSIM dataset. The slopes of these shockwaves range from 16 km/hr to 23 km/hr and, as such, the average backward wave speed is set at 19 km/hr. The jam density is set as 113 veh/lane/km, and a cruising speed of 48 km/hr is used.

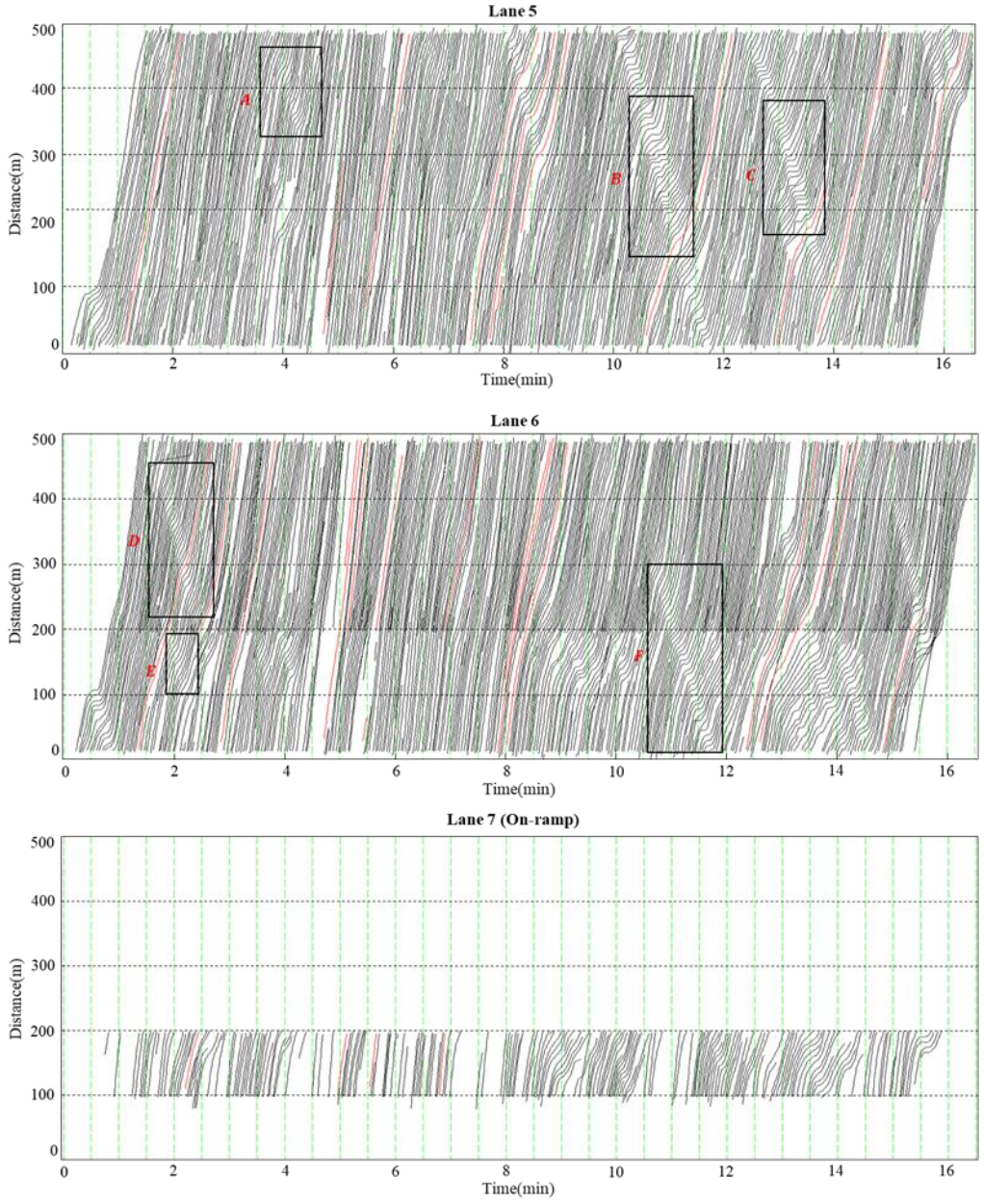


Figure 11. Vehicle trajectory for lanes 5 and 6, and lane 7 (on-ramp), 04:00-04:15, I-80, NGSIM.

According to Eq.(1), the maximum discharge rate of the mixed traffic can be calculated as $\mu_m = \frac{2k_{j,m} \cdot v_u \cdot w_m}{v_u + w_m} = 3{,}086$ veh/hr. The mixed-traffic arrival rate on the main road is captured at 122m (or 400ft) upstream of the on-ramp merging area. The number of vehicles arriving on lane 5, lane 6, and the on-ramp lane 7 are 302, 265, and 192, respectively, in 15 minutes. As such, the arrival rate $\lambda(t)$ is $(302 + 265 +$

192) × 4 = 3,036 veh/hr. The vehicles on the ramp can accelerate on the acceleration lane or lane 6 to reach the full cruising speed. The length of the acceleration lane is 43 m, and an additional distance of 213 m is taken to determine the bottleneck downstream location, so that vehicles are fully accelerated at this location.

NGSIM data shows that, in 15 minutes, the number of vehicles that pass the downstream location of lane 5 and lane 6 are 299 and 280, respectively. As such, the ground truth discharge rate is $(299 + 280) * 4 = 2{,}316$ veh/hr. Analysis of the NGSIM on-ramp merging vehicle trajectory data shows that their average acceleration values are approximately 1.5 $\mathrm{m/s^2}$, i.e. $a_g = 1.5\ \mathrm{m/s^2}$. As the initial speed of merging vehicles on the freeway is not 0, the discount factor is $\frac{v_u \cdot \lambda(t) \cdot \rho_e(t)}{2a_g} \cdot \varphi_2$ based on Eq.(30). The expectation of an effective discharge rate can be calculated as $E(\mu_m{}') = 3086 \times \left(1 - \frac{30 \times 192 \times 4}{2 \times 5 \times \frac{3600^2}{5280}} \times \left[\frac{12}{30} \cdot (\frac{1}{2} + \frac{12}{30}) - \left(\frac{12}{30}\right)^2 \cdot \left(1 + \frac{12}{30}\right) \cdot \ln\left(1 + \frac{30}{12}\right)\right]\right) = 2{,}856$ veh/hr. Compared with the ground truth discharge rate, the absolute percentage error (APE) can be calculated as $APE_1 = \frac{|2{,}856 - 2316|}{2{,}316} = 23.3\%$.

On the other hand, if we ignore the impact of lane-changers from the on-ramp and directly use the maximum discharge rate μ_m, a conclusion of free-flow condition might be obtained, as the maximum discharge rate 3,036 is higher than the traffic flow rate of 2,316, which is certainly not the case. In other words, if an effective discharge rate is not correctly modeled, the APE will become $APE_2 = \frac{|3086 - 2316|}{2316} = 33.2\%$.

We repeat the above process to calculate the *APE* for the other two datasets from I-80. Between 5:00 pm and 5:15 pm, the *APE* can be calculated as $APE_1 = 12.4\%$ and $APE_2 = 20.8\%$. Between 05:15 pm and 5:30 pm, the *APE* can be calculated as $APE_1 = 10.6\%$ and $APE_2 = 19.4\%$. Comparison of APE_1 and APE_2 suggests that, using the proposed effective discharge rate is more reasonable than directly using the maximum discharge rate, and the resulting APE values all drop in the three 15-min time periods.

6.2 Optimal bus-only lane scheduling results analysis

In this section, we show the results of bus-only lane scheduling optimization. Figure 12 presents the demand setup, in which a total of 3 hours is analyzed. The demand starts with a lower value $\lambda_{\mathrm{off-peak}} = 1{,}000$ veh/hr in the first hour, sharply increases to $\lambda_{\mathrm{peak}} = 2{,}500$ veh/hr during the second hour, and then reduces back to $\lambda_{\mathrm{off-peak}} = 1{,}000$ veh/hr. Note that we follow the previously defined notations, and have $t_s = 0, t_0 = 1, t_2 = 2, t_e = 3$. The vehicle ratios are set as $\rho_g(t) = 0.8$, $\rho_b(t) = 0.15$ and $\rho_e(t) = 0.05$ during peak hours, while that are set as $\rho_g(t) = 0.75$, $\rho_b(t) = 0.2$ and $\rho_e(t) = 0.05$ during non-peak hours. The cruising speeds are set at below $v_g = 65$ km/hr(40 mph), $v_b = 40$ km/hr (25 mph) $v_u = 55$ km/hr (35 mph) and the backward wave speed is $w = 19$ km/hr (12 mph). The jam densities are set at below $k_{j,g} = 112$ veh/lane/km (180 veh/lane/mile), $k_{j,b} = 62$ veh/lane/km (80 buses/lane/mile) and $k_{j,m} = 75$ veh/lane/km (140 veh/lane/mile). For the lane-changer, the acceleration value is set at $a_g = 1.5\ \mathrm{m/s^2}$.

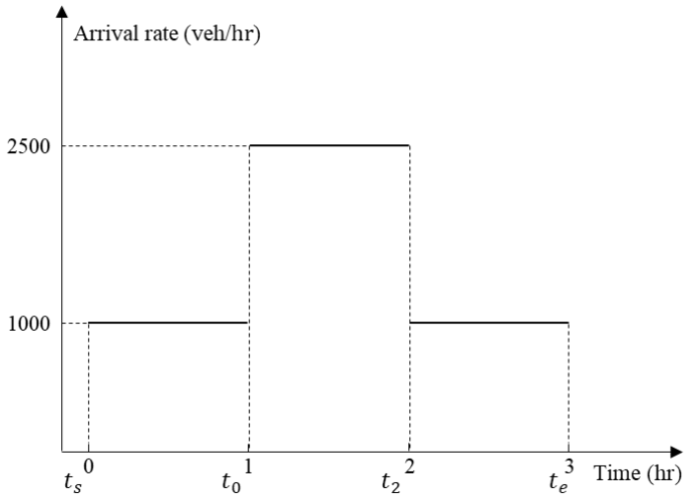


Figure 12. Time-dependent Demand pattern illustration.

Next, we calculate and compare the traffic delays resulting from the demand-oriented strategy and the proposed model. Here, demand-oriented strategy means that the bus-only lane is only activated during peak hours, and the resulting delay is calculated via Eq. (32).

6.2.1 Resulting delay with demand-oriented strategy

Following the demand-oriented strategy, we have $t_{s_2,s} = t_0 = 1$ and $t_{s_2,e} = t_2 = 2$. We denote the resulting delay with a maximum discharge rate μ as $W^0(t_{s_2,s}, t_{s_2,e})$, and with a derived effective discharge rate of $W^1(t_{s_2,s}, t_{s_2,e})$.

According to Eq. (1), the maximum discharge rates are $\mu_m = 1{,}072$ veh/hr/lane, $\mu_g = 1{,}662$ veh/hr/lane and $\mu_b = 811$ veh/hr/lane. For the arterial scenario, the initial speed is $v_0 = 0$. According to Eqs.(11), (19), and(30), the expected discharge rate can be calculated as $E[\mu_m{}'] = 1{,}050$ veh/hr/lane, $E(\mu_g{}') = 1{,}594$ veh/hr/lane and $E(\mu_b{}') = 783$ veh/hr/lane during off-peak hours. During the peak hour, the expected effective discharge rate can be calculated as $E[\mu_m{}'] = 1{,}017$ veh/hr/lane, $E(\mu_g{}') = 1{,}491$veh/hr/lane and $E(\mu_b{}') = 742$ veh/hr/lane. For the freeway scenario, the initial speed is $v_0 \neq 0$. According to Eqs.(11), (19), and(30), the expected discharge rate can be calculated as $E[\mu_m{}'] = 1{,}066$ veh/hr/lane, $E(\mu_g{}') = 1{,}621$ veh/hr/lane and $E(\mu_b{}') = 802$ veh/hr/lane during the off-peak hours. During the peak hour, the expected effective discharge rate can be calculated as $E[\mu_m{}'] = 1{,}058$ veh/hr/lane, $E(\mu_g{}') = 1{,}560$ veh/hr/lane and $E(\mu_b{}') = 788$ veh/hr/lane.

According to Eqs. (32)~(37), the total delay resulting from the demand-oriented strategy with the maximum discharge rate is $W^0(t_{s_2,s}, t_{s_2,e}) = \int_{t_s}^{t_{s_2,s}} Q_{s_1,1}(t)dt + \int_{t_{s_2,s}}^{t_{s_2,e}} Q_{s_2}(t)dt + \int_{t_{s_2,e}}^{t_e} Q_{s_1,2}(t)dt = 325$ veh-hr

Then, we repeat this process with the derived effective discharge rate. For the arterial scenario, the total delay resulting from demand-oriented strategy with an effective discharge rate is: $W^1(t_{s_2,s}, t_{s_2,e}) = \int_{t_s}^{t_{s_2,s}} Q_{s_1,1}(t)dt + \int_{t_{s_2,s}}^{t_{s_2,e}} Q_{s_2}(t)dt + \int_{t_{s_2,e}}^{t_e} Q_{s_1,2}(t)dt = 500$ veh-hr. For the freeway scenario, the total delay resulting from the demand-oriented strategy with an effective discharge rate is: $W^1(t_{s_2,s}, t_{s_2,e}) = \int_{t_s}^{t_{s_2,s}} Q_{s_1,1}(t)dt + \int_{t_{s_2,s}}^{t_{s_2,e}} Q_{s_2}(t)dt + \int_{t_{s_2,e}}^{t_e} Q_{s_1,2}(t)dt = 423$ veh-hr.

6.2.2 Resulting delay with optimized bus-only lane activation schedule

To determine the optimal bus-only lane schedule, four possible cases are considered below, depending on whether the bus-only lane is activated before the peak hour starts, and whether it is terminated after the peak hour ends.

Case 1: $t_s \le t_{s_2,s} \le t_0$ and $t_0 \le t_{s_2,e} \le t_2$

In Case 1, the bus-only lane is activated before the peak hour starts, and deactivated before the peak hour ends. The minimum total delay, with the maximum discharge rate used, is $W^0(t_{s_2,s}, t_{s_2,e}) = 233$ veh-hr and the optimal strategy is not to activate the bus-only lane. When the effective discharge rate is used, for the arterial scenario, the minimum total delay is $W^1(t_{s_2,s}, t_{s_2,e}) = 331$ veh-hr and the optimal strategy is not to activate the bus-only lane. For the freeway scenario, the minimum total delay is $W^1(t_{s_2,s}, t_{s_2,e}) =$ 257 veh-hr and the optimal strategy is not to activate the bus-only lane. For the sake of simplicity, a detailed calculation is omitted.

Case 2: $t_s \le t_{s_2,s} \le t_0$ and $t_2 < t_{s_2,e} \le t_e$

In Case 2, the bus-only lane is activated before the peak hour starts, and deactivated after the peak hour ends. The minimum total delay with the maximum discharge rate can be calculated $W^0\left(t_{s_2,s}, t_{s_2,e}\right) = 356$ veh-hr and an optimal bus-only lane time window of [(0~1), (2~3)]. When the effective discharge rate is used, for the arterial scenario, the minimum total delay is $W^1\left(t_{s_2,s}, t_{s_2,e}\right) =$500 veh-hr and the optimal bus-only lane time window is [(0~1), 2]. For the freeway scenario, the minimum total delay, with maximum discharge rate, can be calculated $W^0\left(t_{s_2,s}, t_{s_2,e}\right) = 477$ veh-hr and an optimal bus-only lane time window of [(0~1), (2~3)].

Case 3: $t_0 \le t_{s_2,s} \le t_2$ and $t_0 \le t_{s_2,e} \le t_2$

In Case 3, the bus-only lane is activated after the peak hour starts, and deactivated before the peak hour ends. The minimum total delay with the maximum discharge rate $is W^0\left(t_{s_2,s}, t_{s_2,e}\right) = 188$ veh-hr and an optimal bus-only lane time window is [1.52, 1.73]. When the effective discharge rate is used, for the arterial scenario, the minimum total delay is $W^1(t_{s_2,s}, t_{s_2,e}) =$285 veh-hr and the optimal bus-only lane time window is [1.52, 1.85]. For the freeway scenario, the minimum total delay, with a maximum discharge rate, can be calculated $W^0\left(t_{s_2,s}, t_{s_2,e}\right) = 219$ veh-hr and an optimal bus-only lane time window of [1.52, 1.77].

Case 4: $t_0 \le t_{s_2,s} \le t_2$ and $t_2 \le t_{s_2,e} \le t_e$

In case 4, the bus-only lane is activated after the peak hour starts, and deactivated also after the peak hour ends. The minimum total delay with maximum discharge rate is $W^0(t_{s_2,s}, t_{s_2,e}) =$204 veh-hr and the optimal bus-only lane time window is [1.7, (2~3)]. When the effective discharge rate is used, at the arterial scenario the minimum total delay is $W^1(t_{s_2,s}, t_{s_2,e}) =$301 veh-hr and the optimal bus-only lane time window is [1.80, (2~3)]. At the freeway scenario, the minimum total delay with maximum discharge rate can be calculated $W^0\left(t_{s_2,s}, t_{s_2,e}\right) = 235$ veh-hr and an optimal bus-only lane time window of [1.83, (2~3)].

Combining the findings from these four cases, the results are summarized in Table 2. These results show that the optimal bus-only lane operation time is [1.52, 1.85] for the arterial scenario and [1.52, 1.77] for the freeway scenario, i.e., activate the bus-only lane after the peak hour starts, and deactivate it at the end of the peak hour period. The resulting lowest total delay is 285 veh-hr for the arterial scenario and 219 veh-hr for the freeway scenario. However, if the effective discharge rate is ignored and the maximum discharge rate is mistakenly used, the decision would be to activate the bus-only lane exactly during [1.52, 1.73], and the delay would be severely underestimated at 188 veh-hr. On the other hand, if a demand-oriented strategy is adopted, the bus-only lane would always be in use exactly at the peak hour, and the resulting delays would be 325, 500, and 423 veh-hr, respectively, with the maximum and effective discharge rates used.

Table 2 Optimized time period of bus-only lane and minimum total delay

Activation time of bus-only lane	Deactivation time of bus-only lane	With maximum discharge rate used		With effective discharge rate used			
				Arterial		Freeway	
$t_{s_2,s}$	$t_{s_2,e}$	$\min W^0$ (veh-hr)	$[t_{s_2,s}, t_{s_2,e}]$	$\min W^1$ (veh-hr)	$[t_{s_2,s}, t_{s_2,e}]$	$\min W^1$ (veh-hr)	$[t_{s_2,s}, t_{s_2,e}]$
$t_s \le t_{s_2,s} \le t_0$	$t_0 \le t_{s_2,e} \le t_2$	233	0	331	0	257	0
$t_s \le t_{s_2,s} \le t_0$	$t_2 < t_{s_2,e} \le t_e$	356	[0~1, 2~3]	500	[0~1, 2]	477	[0~1, 2~3]
$t_0 \le t_{s_2,s} \le t_2$	$t_0 \le t_{s_2,e} \le t_2$	188	[1.52, 1.73]	285	[1.52, 1.85]	219	[1.52, 1.77]
$t_0 \le t_{s_2,s} \le t_2$	$t_2 \le t_{s_2,e} \le t_e$	204	[1.7, 2~3]	301	[1.80, 2~3]	235	[1.81, 2~3]

Two conclusions can be drawn from Table 2. (a) Compared with the demand-oriented strategy that only activates a bus-only lane during peak hours, the proposed bus-only lane scheduling optimization algorithm can reduce traffic delay by (500-285)/500 =43.0% for the arterial scenario, and (423-219)/423 =48.2% for the freeway, and (b) If the effective discharge rate is ignored and the maximum discharge rate is mistakenly used, the bus-only lane operation time will be different and the resulting delay will be underestimated by (285-188)/285 = 34.0% for the arterial road and (219-188)/219 = 14.2% for the freeway.

6.3 Sensitivity analysis

In this section, we perform a sensitivity analysis on an arterial road to show how changing the flow rate and the proportion of vehicles can influence the optimal scheduling of the bus-only lane and the minimum total delay. This sensitivity analysis starts with a set of baseline variables, as follow: $t_s = 0, t_e = 3, t_0 = 1, t_2 = 2, \lambda_{\text{off-peak}} = 1{,}000\,\frac{\text{veh}}{\text{hr}}, \lambda_{\text{peak}} = 2{,}500\,\frac{\text{veh}}{\text{hr}}$. To test what would happen if the ratio of vehicles changes, we set $\rho_g(t) = 0.8$, $\rho_b(t) = 0.15$ and $\rho_e(t) = 0.05$ as benchmarks for both peak hours and non-peak hours.

6.3.1 When flow rate changes

We start by testing to see what would happen if the flow rate λ_{peak} changes. Figure 13, (a) and (b) illustrate the peak hour arrival rate increasing from 2,500 veh/hr to 4,000 veh/hr, whereas (c) and (d) show the off-peak hour arrival rate increasing from 500 veh/hr to 2,000 veh/hr. Also, (a) and (c) in Figure 13 illustrate the time to activate and deactivate the bus-only lane, and (b) and (d) describe the minimum traffic total delay. The notations are defined in the same way, as earlier, that $t^0_{s_2,s}$, $t^0_{s_2,e}$, $W^0_{\min}$ refer to the start and end time of bus-only lane operation, and the total delay with the maximum discharge rate used, while $t^1_{s_2,s}$, $t^1_{s_2,e}$, $W^1_{\min}$ are used with the derived effective discharge rate, all following the proposed optimal bus-only lane schedule. For comparison purposes, W^1_0 is also shown, which is the resulting delay with the derived effective discharge rate, but with the demand-oriented strategy used. $\Delta_1 = \frac{W^1_{\min} - W^0_{\min}}{W^1_{\min}} \cdot 100\%$ is defined to show the percentage of traffic delay underestimation, if the maximum discharge rate is mistakenly used. $\Delta_2 = \frac{W^1_0 - W^1_{\min}}{W^1_0} \cdot 100\%$ is defined to show the percentage of traffic delay reduction with the proposed bus-only lane scheduling optimization algorithm.

As shown in Figure 13, the bus-only lane would be deactivated later as λ_{peak} increases. Figure 13 (a) shows that, when the maximum discharge rate is used, the activation time $t^0_{s_2,s}$ advances from 1.45 hr to 1.14 hr and the deactivation time is stable at the end of the peak hour. When the effective discharge rate is used,

the activation time $t^1_{s_2,s}$ is stable at around 1.5 hr and the deactivation time $t^1_{s_2,e}$ postpones from 1.85 hr to 2.65 hr. In Figure 13 (b), when λ_{peak} increases, the three delay curves all go up. The difference in increases between line W^1_0 and $W^1_{\min}$ also suggests that, when demand increases, the proposed bus-only lane optimization algorithm can achieve a more significant benefit. In terms of traffic delay savings percentage, by observing Δ_2, it can be found that its value ranges from 43% to 7 %, with an average of 23%, meaning that the total delay drops 23%, on average, if the bus-only lane is activated following the optimal scheduling algorithm, instead of the demand-oriented strategy. On the other hand, Δ_1 ranges from 40% to 20%, with an average value of 31%, which means that the total delay would have been underestimated by 31% if the maximum discharge rate is mistakenly used.

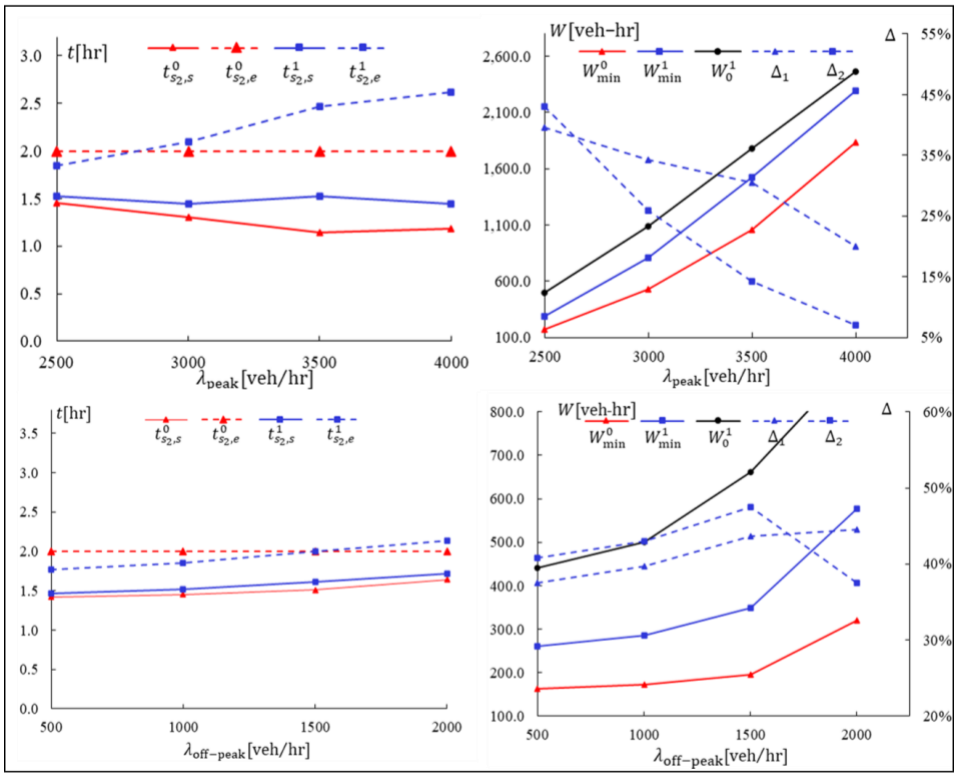


Figure 13. Optimal period for bus-only lane and minimal total delay when the arrival rate changes.

Figure 13 (c) shows that, when the arrival rate during the off-peak hour increases, the time to activate the bus-only lane is slightly postponed. The time to deactivate is stable, when using the maximum discharge rate, and it deactivates later, when using the effective discharge rate. Similarly, when $\lambda_{\text{off-peak}}$ increases, the three delay curves all go up. The difference between line W^1_0 and $W^1_{\min}$ also increases, suggesting that, when the off-peak hour demand increases, the proposed bus-only lane optimization algorithm can also achieve a more significant benefit. In terms of a traffic delay savings percentage, by observing Δ_2, it can be found that its value ranges from 37% to 41%, with an average of 42%, meaning that the total delay drops 42% on averages if the bus-only lane is activated following the optimal scheduling algorithms instead of the demand-oriented strategy. On the other hand, Δ_1 ranges from 38% to 45%, with an average value of 41%, which means that the total delay would have been underestimated by 41% if the maximum discharge rate had been mistakenly used.

6.3.2 When the ratio of vehicles changes

We now examine what would happen if the vehicle ratios $\rho_g(t)$, $\rho_b(t)$, and $\rho_e(t)$ change (shown in Figure 14). We fix $\rho_b(t) = 0.15$, and change $\rho_g(t)$ and $\rho_e(t)$ in (a) and (b), whereas we fix $\rho_e(t) = 0.05$, and

change $\rho_g(t)$ and $\rho_b(t)$ in (c) and (d), and we fix $\rho_g(t) = 0.8$, and change $\rho_b(t)$ and $\rho_e(t)$ in (e) and (f). The notations of $t^0_{s_2,s}$, $t^0_{s_2,e}$, $W^0_{\min}$, $t^1_{s_2,s}$, $t^1_{s_2,e}$, $W^1_{\min}$, W^1_0, Δ_1 and Δ_2 are the same, and are not repeated here.

Figure 14 (a) shows that, when the proportion of buses is fixed at 0.15, as $\rho_e(t)$ increases from 0.01 to 0.1, the $\rho_g(t)$ decreases from 0.84 to 0.75. When the effective discharge rate is used, the time to activate a bus-only lane remains practically unchanged, and the time to terminate postpones from 1.77 hr to 2.26 hr. The period for the bus-only lane increases from 0.25 hr to 0.48 hr. However, if the maximum discharge rate is used, the time to activate or to deactivate a bus-only lane is not at all sensitive. Figure 14 (b) shows that, as the lane-changer traffic increases, the two total delay curves using the effective discharge rate increase. The difference between line W^1_0 and $W^1_{\min}$ increases, suggesting that, when the lane-changer traffic increases, the proposed bus-only lane optimization algorithm can achieve a more significant benefit. In terms of a traffic delay savings percentage, by observing Δ_2, it can be found that its value is stable around 43%, meaning that the total delay drops 43%, on average, if the bus-only lane is activated following the optimal scheduling algorithm, instead of the demand-oriented strategy. On the other hand, Δ_1 increases sharply from, 8% to 54%, for a 34% average value, which means that the total delay would be underestimated by 34% if the maximum discharge rate is mistakenly used.

Figure 14 (c) shows that when the proportion of general vehicles is fixed, the lane-changer ratio increases. When the effective discharge rate is used, the time to activate a bus-only lane is stable, around 1.5 hr at first, and then increases to 2 hr, while the time to terminate is stable around 1.8 hr, and then increases to 2.5 hr. The period for the bus-only lane increases from 0.29 hr to 0.5 hr. However, if the maximum discharge rate is used, the time to activate or to deactivate a bus-only lane is not at all sensitive. Figure 14 (d) shows that, as the lane-changer traffic increases, three total delay curves, using the effective discharge rate, increase. The difference between line W^1_0 and $W^1_{\min}$ increases sharply, suggesting that, when the lane-changer traffic increases, the proposed bus-only lane optimization algorithm can achieve a more significant benefit. In terms of a traffic delay savings percentage, by observing Δ_2, it can be found that the value increases from 28% to 60%, meaning that the total delay drops 47%, on average, if the bus-only lane is activated following the optimal scheduling algorithm, instead of the demand-oriented strategy. On the other hand, Δ_1 increases sharply from 8% to 63%, for a 42% average value, which means that the total delay would be underestimated by 42% if the maximum discharge rate is mistakenly used. Figure 14 (e) shows that, when the proportion of lane-changers is fixed, the bus ratio increases. When the effective discharge rate is used, the time to activate and terminate is stable. The period for the bus-only lane increases from 0.24 hr to 0.76 hr, as the bus ratio increases. However, if the maximum discharge rate is used, the time to activate or to deactivate a bus-only lane fluctuates slightly until the bus ratio reaches a high point (e.g.,19%). Figure 14 (f) shows that, as the ratio of buses increases, three total delay curves, using the effective discharge rate, drop. The difference between line W^1_0 and $W^1_{\min}$ drops sharply, suggesting that, when the ratio of buses increases, the proposed bus-only lane optimization algorithm can achieve a lesser benefit. In terms of a traffic delay savings percentage, by observing Δ_2, it can be found that the value drops from 62% to 9%, meaning that the total delay drops 40%, on average, if the bus-only lane is activated following the optimal scheduling algorithm, instead of the demand-oriented strategy. On the other hand, Δ_1 increases sharply, from 33% to 47%, for a 36% average value, which means that the total delay would be underestimated by 36%, if the maximum discharge rate is mistakenly used.

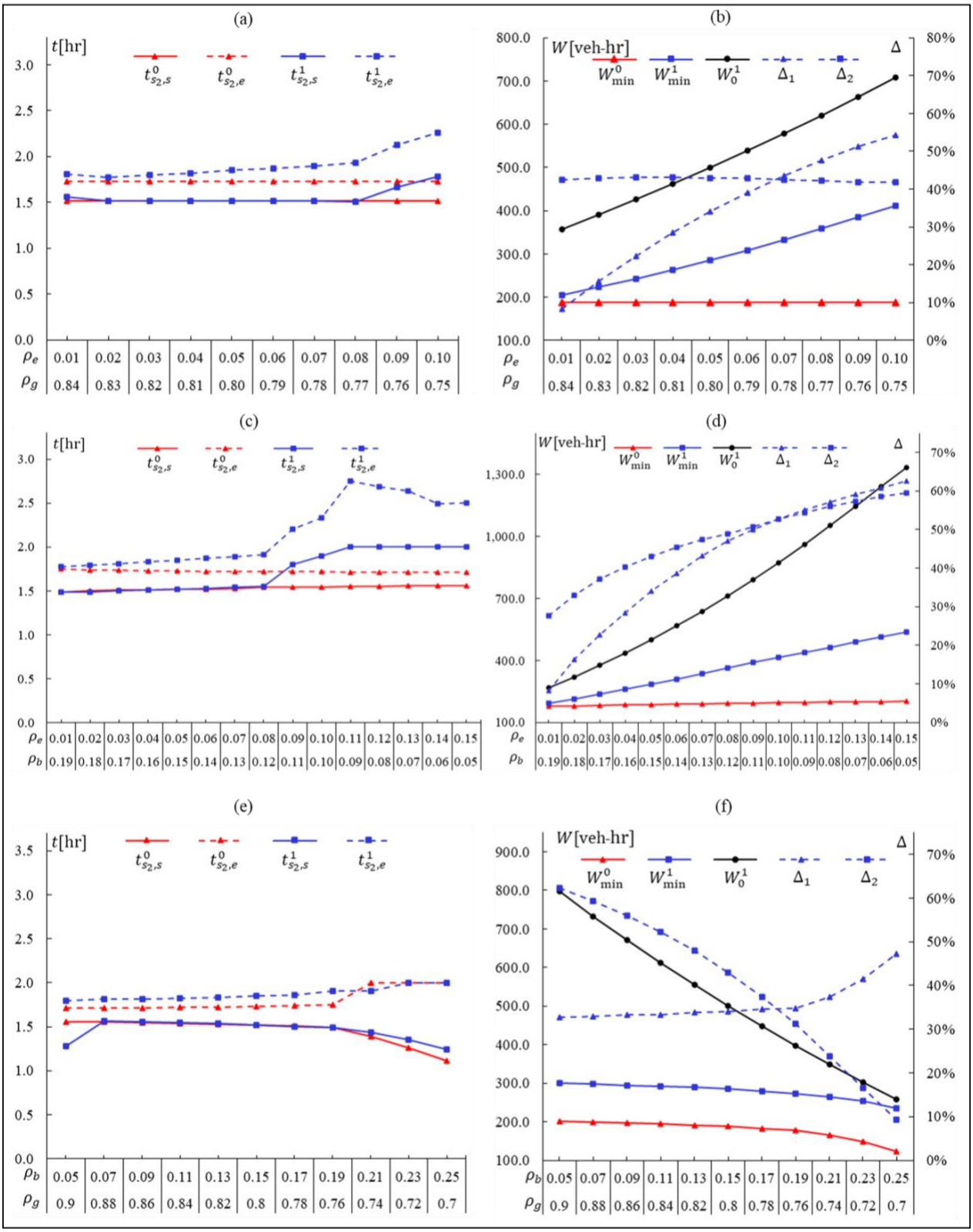


Figure 14. Optimized period for a bus-only lane and minimal total delay when the $\rho(t)$ changes in the arterial scenario.

7. Conclusions and Future Work

This manuscript analytically derives the effective discharge rate of a roadway segment, for the purpose of optimal bus-only lane conversion scheduling. Three effective discharge rate discount factors are analytically derived, including one for bus-only lane, and one for general-purpose lane in a scenario with the bus-only lane activated, and the last one for mixed-traffic lane in a scenario with the bus-only lane deactivated. The formulation is based on multi-stage merging and lane-changing behavior, microscopic car-following behavior, shockwave propagation, and fundamental diagram. The derived effective discharge rates are represented with mathematically-simple expressions in a closed-form, and correctly account for the effects of demand inputs (such as arrival rate and the ratios of vehicles of different types), vehicular performance (such as acceleration rate), and traffic flow characteristics (such as backward wave speed and jam density). With the derived equations, as long as the entering vehicle's demand information $\lambda(t)$ and ρ_e, and the traffic flow property w_m, v_u, a_g are given (which are usually not difficult to obtain in reality), the discount factors for each scenario can be easily calculated.

The optimization of bus-only lane conversion scheduling is then formulated to minimize the total traffic delay. Validation with NGSIM data shows that our model correctly captures the effective discharge rate discount in a mixed traffic flow, resulting in an estimation error reduction from 15.0% to 6.6% on the arterial road, and from 33.2%, 20.8%, and 19.4% to 23.3%, 12.4%, and 10.6% for the three time periods on freeway, noting a significant improvement. It is also suggested that, when compared with the traditional strategy of enabling a bus-only lane during peak hours, the proposed optimization scheduling algorithm can reduce total traffic delay by 43.0% for the arterial scenario, and 48.2% for the freeway. The research also suggests that, if the maximum discharge rate is mistakenly used, the resulting traffic delay will be severely underestimated. Sensitivity analysis is also performed and reveals that the optimal bus-only lane scheduling time varies in responding to the traffic demand and vehicle ratios, and that the developed optimization algorithm is always more beneficial, as compared with the demand-oriented strategy.

In terms of future work, the modeling work to derive effective discharge rate discount factors may be directly applied in a much broader scope, as the roadways' discharge rate (capacity) reduction due to merging traffic is a fundamental problem that commonly occurs. For example, when evaluating the performance of a HOV lane, a key challenge is the capacity drop at the lane entrance/exit locations, where merging behaviors are observed and reduce the effective discharge rate. Another application might be the intelligent merging strategy of connected and autonomous vehicles, so that the traffic on the main road can be interrupted to a lesser degree, and with a lower speed fluctuation, which may also help reduce vehicle crashes at these merging locations.

8. Appendices

Appendix A - Notation List

Car-following model:	
t	timestamp
$n, n-1$	indexes of following and leading vehicle, respectively
τ_n, d_n	response time and temporal delay of vehicle n
v_1, v_2	speeds before and after the speed changes
$h_{n,1}, h_{n,2}$	time headway of vehicle n before and after the speed changes
$s_{n,1}, s_{n,2}$	space headway of vehicle n before and after the speed changes
Link q-k relationship:	
q, k, v	traffic flow, density and speed

$\bar{\tau}, \bar{d}$	arithmetic average of the τ and d
v_f, k_j	free-flow speed and jam density
v_b, v_g, v_u	cruising speed for the bus-only lane, general-purpose lane, and mixed traffic lane. Due to bottleneck or congestion, $v_b, v_g, v_u \leq v_f$
v_e	entering speed of the merging vehicles
w	backward wave speed
μ	maximum discharge rate
Demand-supply and queuing profile:	
$\lambda(t)$	time-dependent arrival rate
t_0, t_1	time that queue starts, and time to reach the maximum arrival rate
$t_2, \bar{t}$	time with longest queue, and time when the queue is fully discharged
$Q(t)$	time-dependent queue length
v_μ	speed when the queue is being discharged
$W(t)$	time-dependent total delay
Modeling of effective discharge rate:	
b, g, e	buses and general vehicles on the main road, and entering vehicles (i.e., lane-changer) that merge into the main road from side-street entrance
m	mixed traffic on the main road when the bus-only lane is deactivated
S_1, S_2	scenario without bus-only lane and scenario with bus-only lane activated
$\lambda_b, \lambda_g, \lambda_e$	flow rates of the buses, general vehicles and entering vehicles
$\rho_b(t), \rho_g(t), \rho_e(t)$	time-dependent ratios of the buses, general vehicles and entering vehicles
h_e	time headway of the entering vehicles
a_g	acceleration of the general vehicles and entering vehicles
w_b, w_g, w_m	backward wave speed of the bus-only lane and general-purpose lane in scenario S_2, and backward wave speed of mixed-traffic lane in scenario S_1
$k_{j,b}, k_{j,g}, k_{j,m}$	jam density of the bus-only lane, general-purpose lane, and mixed-traffic lane
μ_b, μ_g, μ_m	maximum discharge rate of the bus-only lane, general-purpose lane, and mixed-traffic lane
$\mu_b', E[\mu_b'(t)]$	effective discharge rate and its expectation in the bus-only lane
$\mu_g', E[\mu_g'(t)]$	effective discharge rate and its expectation in the general-purpose lane
$\mu_m', E[\mu_m'(t)]$	effective discharge rate and its expectation in the mixed-traffic lane
$v^*(t)$	speed of the entering vehicle before it reaches full cruising speed
$x^*(t)$	location of a moving observer relative to the entering vehicle
$q^*(t)$	flow rate at which vehicle pass the observer who is travelling at a speed of $v^*(t)$

$Q(k), Q^*(k)$	flow as a function of density seen from stationary and moving coordinates
$\lambda_{\text{in}}(t), \lambda'_{\text{in}}(t)$	adjusted flow rate from stationary coordinate and moving coordinate
t_s, t_e	start and end time of the analysis period
$t_{s_2,s}, t_{s_2,e}$	start and end time of scenario S_2
$Q_{s_1}(t)$	time-dependent queue length of scenario S_1
$Q_{s_2,b}(t), Q_{s_2,g}(t)$	time-dependent queue length of bus-only lane and general-purpose lane of S_2
$\lambda_{\text{peak}}, \lambda_{\text{off-peak}}$	flow rate during the peak hour and off-peak hour
APE	absolute percentage error
$W^0_{\min}$	minimal total delay without considering the effective discharge rate
$W^1_{\min}$	minimal total delay considering the effective discharge rate
W^1_0	total delay when implementing the bus-only lane during the peak-hour $t \in [t_{\text{peak},s}, t_{\text{peak},e}]$

Appendix B – Some Derivations

Derivation of bus-only lane effective discharge rate expression in Eq. (10)

$$\mu'_b = \frac{\mu_{BE'} \cdot t_{BE'} + \mu_b(t_{AB} + t_{E'F})}{h_e}$$

$$= \frac{\frac{w_b \cdot k_{j,b} \cdot (v_{c'} + v_e)}{v_{c'} + 2w_b + v_e} \cdot \left(\frac{v_{C'}^2 - v_e^2}{2a_g w_b} + \frac{v_{C'} - v_e}{a_g}\right) + \frac{k_{j,b} \cdot v_b \cdot w_b}{v_b + w_b} \cdot \left(\frac{1}{\lambda(t) \cdot \rho_e(t)} - \left(\frac{v_{C'}^2 - v_e^2}{2a_g w_b} + \frac{v_{C'} - v_e}{a_g}\right)\right)}{\frac{1}{\lambda(t) \cdot \rho_e(t)}}$$

$$= \lambda(t) \cdot \rho_e(t) \cdot \left(\frac{k_{j,b} \cdot \left(v_{C'}^2 - v_e^2\right)}{2a_g} + \frac{k_{j,b} \cdot v_b \cdot w_b}{(v_b + w_b) \cdot \lambda(t) \cdot \rho_e(t)} - \frac{k_{j,b} \cdot v_u \cdot (v_{c'} - v_e) \cdot (v_{c'} + 2w_b + v_e)}{2a_g \cdot (v_u + w_b)}\right)$$

$$= \frac{k_{j,b} \cdot v_b \cdot w_b}{v_b + w_b} \cdot \left[1 - \frac{\lambda(t) \cdot \rho_e(t)}{2a_g} \cdot \left(2(v_{C'} - v_e) - \frac{v_{C'}^2 - v_e^2}{v_b}\right)\right]$$

$$= \mu_b \cdot \left[1 - \lambda(t) \cdot \rho_e(t) \cdot \left(\frac{v_{C'} - v_e}{a_g} - \frac{v_{C'}^2 - v_e^2}{2a_g \cdot v_b}\right)\right]$$

Derivation of general-purpose lane effective discharge rate expression in Eq. (18)

$$\mu_g{}' = \frac{\mu_{C'E} \cdot t_{C'E} + \mu_g \cdot (t_{AC'} + t_{EF})}{h_e}$$

$$= \frac{w_g \cdot k_{j,g} \cdot \frac{v_g + v_{C'}}{v_g + v_{C'} + 2w_g} \cdot \frac{v_g - v_{C'}}{2a_g w_g} \cdot (v_g + v_{C'} + 2w_g)}{\frac{1}{\lambda(t) \cdot \rho_e(t)}} + \frac{k_{j,g} \cdot v_g \cdot w_g}{v_g + w_g}$$

$$\cdot \frac{\left(\frac{1}{\lambda(t) \cdot \rho_e(t)} - \frac{v_g - v_{C'}}{2a_g w_g} \cdot \left(v_g + v_{C'} + 2w_g\right)\right)}{\frac{1}{\lambda(t) \cdot \rho_e(t)}}$$

$$= \lambda(t) \cdot \rho_e(t)$$

$$\cdot \left(\frac{k_j, g}{2a_g} \cdot \left({v_g}^2 - {v_{C'}}^2\right) + \frac{k_{j,g} \cdot v_g \cdot w_g}{\left(v_g + w_g\right) \cdot \lambda(t) \cdot \rho_e(t)}\right.$$

$$\left. - \frac{k_{j,g} \cdot v_g \cdot \left(v_g - v_{C'}\right) \cdot \left(v_g + v_{C'} + 2w_g\right)}{2a_g \cdot \left(v_g + w_g\right)}\right)$$

$$= \frac{k_{j,g} \cdot v_g \cdot w_g}{\left(v_g + w_g\right)} - \lambda(t) \cdot \rho_e(t) \cdot \left(\frac{k_{j,g} \cdot w_g \cdot \left(v_g - v_{C'}\right)^2}{2a_g \cdot \left(v_g + w_g\right)}\right)$$

$$= \frac{k_{j,g} \cdot v_g \cdot w_g}{v_g + w_g} \cdot \left(1 - \frac{\left(v_g - v_{C'}\right)^2 \cdot \lambda(t) \cdot \rho_e(t)}{2a_g \cdot v_g}\right)$$

$$= \mu_g \cdot \left(1 - \frac{\left(v_g - v_{C'}\right)^2 \cdot \lambda(t) \cdot \rho_e(t)}{2a_g \cdot v_g}\right)$$

Derivation of the maximum discharge rate of point H' in the stationary coordinates, i.e., the length of $H'M'$ in Eq. (26).

$$H'M' = HM' + kv^*(t) = \frac{k_{j,m} \cdot w_m \cdot \left(v_u - v^*(t)\right)}{2(v_u + w_m)} + \frac{k_{j,m} \cdot w_m \cdot v^*(t)}{2}\left(\frac{1}{v_u + w_m} + \frac{1}{v^*(t) + w_m}\right)$$

$$= \frac{k_{j,m} \cdot v_u \cdot w_m}{v_u + w_m} \cdot \frac{2v^*(t) \cdot v_u + w_m \cdot v_u + w_m \cdot v^*(t)}{2(v^*(t) + w_m) \cdot v_u}$$

$$= \mu_m \cdot \frac{2v_u \cdot v^*(t) + v^*(t) \cdot w_m + w_m \cdot v_u}{2(v^*(t) + w_m) \cdot v_u}$$

Derivation of the effective discharge rate of the moving bottleneck in arterial scenario with $v_e = 0$, between time $[0, \frac{v_u}{a_g}]$, in Eq. (27).

$$E_{\left[0,\frac{v_u}{a_g}\right]}[\mu_m{}'(t)] = E[\mu_m \cdot \frac{(2v_u \cdot v^*(t) + v^*(t) \cdot w_m + w_m \cdot v_u)}{2(v^*(t) + w_m) \cdot v_u}]$$

$$= \frac{\mu_m}{\frac{v_u}{a_g}} \cdot [\left(1 + \frac{w_m}{2v_u}\right)$$

$$\cdot \int_0^{\frac{v_u}{a_g}} \left(1 - \frac{w_m}{a_g t + w_m}\right) dt + \frac{w_m}{2} \cdot \int_0^{\frac{v_u}{a_g}} \frac{1}{a_g t + w_m} dt]]$$

$$= \frac{\mu_m \cdot a_g}{v_u} \cdot (\frac{v_u}{a_g} + \frac{w_m}{2a_g} - \frac{w_m}{2a_g} \cdot \ln(v_u + w_m) - \frac{{w_m}^2}{2a_g \cdot v_u} \cdot \ln(v_u + w) + \frac{w_m}{2a_g} \cdot \ln(w_m)$$

$$+ \frac{{w_m}^2}{2a_g \cdot v_u} \cdot \ln(w_m)) = \frac{\mu_m \cdot a_g}{v_u} \cdot (\frac{v_u}{a_g} + \frac{w_m}{2a_g} + \left(\frac{w_m}{2a_g} + \frac{{w_m}^2}{2a_g \cdot v_u}\right) \cdot \ln(\frac{w_m}{v_u + w_m}))$$

Derivation of the effective discharge rate of the moving bottleneck in the arterial scenario with $v_e = 0$, in the stationary coordinates, in Eq. (28).

$$E[\mu_m{}'(t)] = \frac{(\frac{v_u}{a_g} \cdot E_{\left[0,\frac{v_u}{a_g}\right]}[\mu_m{}'(t)] + \mu_m \cdot (\frac{1}{\lambda(t) \cdot \rho_e(t)} - \frac{v_u}{a_g}))}{\frac{1}{\lambda(t) \cdot \rho_e(t)}}$$

$$= \frac{\mu_m \cdot \left(\frac{1}{\lambda(t) \cdot \rho_e(t)} + \frac{w_m}{2a_g} + \left(\frac{w_m}{2a_g} + \frac{{w_m}^2}{2a_g \cdot v_u}\right) \cdot \ln\left(\frac{w_m}{v_u + w}\right)\right)}{\frac{1}{\lambda(t) \cdot \rho_e(t)}}$$

$$= \mu_m \cdot (1 - \frac{w_m \cdot \lambda(t) \cdot \rho_e(t)}{2a_g} \cdot [\left(1 + \frac{w_m}{v_u}\right) \ln\left(1 + \frac{v_u}{w_m}\right) - 1])$$

Derivation of the effective discharge rate of the moving bottleneck in the freeway scenario with $v_0 \neq 0$, between time $[0,\frac{v_u - v_e}{a_g}]$, in Eq. (27).

$$E_{\left[0,\frac{v_u - v_0}{a_g}\right]}[\mu_m{}'(t)] = E[\mu_m \cdot \frac{(2v_u \cdot v^*(t) + v^*(t) \cdot w_m + w_m \cdot v_u)}{2(v^*(t) + w_m) \cdot v_u}]$$

$$= \frac{\mu_m}{\frac{v_u - v_e}{a_g}}$$

$$\cdot \left[\left(1 + \frac{w_m}{2v_u}\right) \cdot \int_0^{\frac{v_u - v_e}{a_g}} \left(1 - \frac{w_m}{v_e + a_g t + w_m}\right) dt + \frac{w_m}{2} \cdot \int_0^{\frac{v_u - v_e}{a_g}} \frac{1}{v_e + a_g t + w_m} dt\right]$$

$$= \frac{\mu_m a_g}{v_u - v_e} \cdot [\left(1 + \frac{w_m}{2v_u}\right) \cdot \left(\frac{v_u - v_e}{a_g} - \frac{w_m}{a_g} \cdot \ln\frac{v_u + w_m}{v_e + w_m}\right) + \frac{w_m}{2a_g} \cdot \ln\frac{v_u + w_m}{v_e + w_m}]$$

$$= \mu_m \cdot \left[\frac{2v_u + w_m}{2v_u} + \frac{w_m(v_u + w_m)}{2v_u(v_u - v_e)} \ln(\frac{v_e + w_m}{v_u + w_m})\right]$$

Derivation of the effective discharge rate of the moving bottleneck in the freeway scenario with $v_e \neq 0$, in the stationary coordinates, in Eq. (28).

$$E[\mu_m{}'(t)] = \frac{(\frac{v_u - v_e}{a_g} \cdot E_{\left[0,\frac{v_u-v_e}{a_g}\right]}[\mu_m{}'(t)] + \mu_m \cdot (\frac{1}{\lambda(t)\cdot\rho_e(t)} - \frac{v_u - v_e}{a_g}))}{\frac{1}{\lambda(t)\cdot\rho_e(t)}}$$

$$= \frac{\mu_m \cdot E\left\{\frac{v_u - v_e}{a_g} \cdot \left[\frac{2v_u + w_m}{2v_u} + \frac{w_m(v_u + w_m)}{2v_u(v_u - v_e)} \ln\left(\frac{v_e + w_m}{v_u + w_m}\right)\right] + \left(\frac{1}{\lambda(t)\cdot\rho_e(t)} - \frac{v_u - v_e}{a_g}\right)\right\}}{\frac{1}{\lambda(t)\cdot\rho_e(t)}}$$

$$= \frac{\mu_m \cdot \left\{\frac{v_u - E(v_e)}{a_g} \cdot \frac{2v_u + w_m}{2v_u} + \frac{w_m(v_u + w_m)}{2v_u a_g} \cdot E[\ln\left(\frac{v_e + w_m}{v_u + w_m}\right)] + \left(\frac{1}{\lambda(t)\cdot\rho_e(t)} - \frac{v_u - E(v_e)}{a_g}\right)\right\}}{\frac{1}{\lambda(t)\cdot\rho_e(t)}}$$

$$= \mu_m \cdot \left\{1 - \frac{v_u\lambda(t)\rho_e(t)}{2a_g} \cdot \left[\frac{w_m}{v_u} \cdot (\frac{1}{2} + \frac{w_m}{v_u}) - \left(\frac{w_m}{v_u}\right)^2 \cdot \left(1 + \frac{w_m}{v_u}\right) \cdot \ln\left(1 + \frac{v_u}{w_m}\right)\right]\right\}$$

9. Statement of Contribution

Qing Tang: Formal analysis, Validation, Visualization, Writing - original draft. Xianbiao Hu: Conceptualization, Methodology, Supervision, Writing - original draft. Xuewei Lu: Validation, Visualization. Jiawei Lu, Xuesong Zhou, Conceptualization, Methodology, Editing - original draft.

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